Liquid intrusion and evaporation in low-pressure cavities: numerical investigation with STAR-CCM+

Einströmung und Verdampfung einer Flüssigkeit unter Niedrigdruckbedingungen: numerische Simulationen mit STAR-CCM+

To use the RWTH logo on your title page, you need to submit a special form with your thesis!

Masterarbeit

Master's Thesis

presented by / von

Pandey, Harsh

415518

Supervisors / Betreuer:

Benjamin Terschanski

Prof. Dr. Julia Kowalski

Aachen, *12th September, 2025*

Contents

1 Introduction **1**
- 1.1 The Problem . . . 1
- 1.2 Multiphase Flows and Star-CCM+ . . . 2
- 1.3 Objectives and Outline . . . 2

2 Modelling the Physical Problem **4**
- 2.1 Evaporation: Phenomenological Overview and Literature Review . . . 4
  - 2.1.1 Effects of low pressure and gravity conditions . . . 4
  - 2.1.2 Vaporisation in a superheated liquid . . . 5
- 2.2 Modelling Multiphase Flows . . . 5
  - 2.2.1 The One-Fluid Formulation . . . 7
  - 2.2.2 The Two-Fluid Formulation . . . 8
- 2.3 Modelling Turbulence . . . 9
- 2.4 Modelling Subproblem 1 . . . 10
- 2.5 Modelling Subproblem 2 . . . 10

3 Nums **11**
- 3.1 VOF . . . 11
- 3.2 EMP . . . 11
- 3.3 Star-CCM+ Stuff - needs a better heading . . . 12
- 3.4 Momentum Closures . . . 12
- 3.5 Closures: Under Construction . . . 14
- 3.6 Vague and Miscellaneous . . . 14

4 Computational Methodology and Results **15**
- 4.1 Stefan Problem . . . 15
- 4.2 Rayleigh-Taylor Instability . . . 18
- 4.3 Liquid Infiltration of Evaporation Chamber . . . 24

References **27**

1 Introduction

Their potential to host extraterrestrial life has made icy moons, such as the Jovian moon Europa and Saturn's moon Enceladus, prime targets for exploration in the upcoming decades. The Cassini-Huygens mission discovered active cryovolcanism on Enceladaus in 2005 in the form of water rich plumes erupting from the south polar region. These plumes consist of [what and what], and supply most of the material making up Saturn's E ring. [sklaventis, fontes]

The mechanisms of cryovolcanism and the disparity between the recorded flow behaviour and contituents of the plumes are subject to ongoing research. Various models aim to explain the plume mechanics and the observed flow behaviour - explosive boiling due to exposure of the subsurface ocean to vacuum due to cracks opened by tidal stresses; simulating the fissures in the surface as convergent-divergent (de Laval) nozzles that accelerate the vapor to the observed supersonic speeds; or channels inside the icy crust acting as a de Laval nozzle where the grains are initially decelerated due to frequent wall collisions before being re-accelerated by the vapour flow. [sklaventis, fontes, porco, Brilliantov & Schmidt (2016), yeoh et al 2016] To be able to use plumes as a surrogate to learn about the subsurface ocean, a better understanding of their driving processes in key.

As part of the icy moon experiments, in an effort to contribute to the understanding of the fluid dynamics of the supersonic plumes, an experimental setup is under development at the FH Aachen - University of Applied Sciences. In collaboration with the Rocket Experiments for University Students (REXUS) programme of the German Aerospace Center (DLR) and the Swedish Space Agency (SNSA), the experimental setup will be installed on a sounding rocket and will perform under reduced gravity and vacuum conditions. The experiment consists of heated liquid water being let into a chamber connected to a de Laval nozzle, initially filled with low-pressure air. The nozzle lid is opened after a short time period and this depressurisation of the chamber is expected to result in vaporisation of the water present in the evaporation chamber.

1.1 The Problem

The 'flow system assembly' in the experimental setup, as shown in figure [FIG], consists of the hydraulic accumulator, the evaporation chamber, and the injection system subassembly that drives the liquid from the hydraulic accumulator into the evaporation chamber. In the current design of the setup, these components are assembled vertically and the water is driven up into the evaporation chamber. The evaporation chamber is constructed out of aluminium and can be assumed to be separated into a water reservior and the nozzle. [-dimensions of reservoir, nozzle. CAD in appendix; where are flow measurements made]

The hydraulic accumulator and the evaporation chamber can be heated until lift-off. Current design considerations assume the temperature of the evaporation chamber and water in the accumulator to be 368.15 K and 353.15 K respectively. The ambient temperature at lift-off is assumed to be 280 K based on the remarks and considerations made in [OLD-SED]. The flow system assembly is sealed off before lift-off and an intial pressure of 1 bar is assumed in the evaporation chamber, initially occupied by air. The hydraulic accumulator is sufficiently insulated and the nozzle is encased in a golden foil to minimise the amount of heat lost. No stipulations have been made regarding the evaporation chamber, however as there is no fluid convection during operation, losses are assumed to be minimum.

Water injection begins after the electronics module signals the start of data storage and ends shortly before the apogee is reached. Due to water remaining at the time of cutoff in the injection subassembly and also the accumulator, the total amount of water discharged to the chamber is assumed to be 150 mL. A constant mass flow rate of 15 mL/s is assumed. The inflow velocity in the actual experiment can be tuned to desired values. Discharge time in the simulation is calculated based on the estimated water level at the end of discharge. A minimum time of 6 s and a system pressure of 1.7 bar is assumed before the electronics system signals the opening of the nozzle lid ensuing the vaporisation of the water. As no active heating is provided, the phase change also takes place in the fluid bulk. This process is facilitated by the presence of nucleation points that are found in the water as 2.5 g of silicates per 100 mL of water. The silicates are assumed to be spherical with a uniform diameter of $0.8\,\mu$m. At the start of depressurisation, these 'seed' points are assumed to be uniformly distributed in the water bulk.

TT

1.2 Multiphase Flows and Star-CCM+

Multiphase flows are characterised by the simultaneous flow of several phases. Regardless of our improved understanding of such flows, multiphase flows remain complicated to model. Multiphase flows are multiscale in nature - where the cascasding effects of various physics acting over system-level to the microscopic scale must be taken into account. [1] Along with turbulence, multiple deformable, moving interfaces pose discontinuities in the fluid properties and induce fluctuations in the flow variables in the neighbourhood. The distribution of the phases cannot be accurately estimated a priori [2] but the macroscopic behaviour of the system is dependent on small-scale interactions of the phases [3]. These evolving interfacial structures tend to directly affect the inertial and thermodynamical development of the flow field. [1]

Two-phase flow can be classified into three main flow regimes based on the distribution of the phases - dispersed flow, transitional flow, and separated flow. In case of liquid-gas flows, this ranges from bubbly flows - where small gas bubbles are finely dispersed through the continuous liquid phase, to the separated flows - where a sharp interface marks the boundary between the two phases. The transitional flow regime is marked by the presence of both dispersed and separated flows. [2]

Star-CCM+ is a commerical CFD software currently maintained and developed by Siemens Digital Industries Software. The software uses the Finite Volume Method to discretise the integral form of the governing differential equations over a number of control volumes in the domain. The software has an automatic meshing system but allows the implementation of custom meshing using surface wrappers. The relevant variables are calculated at the center of each control volume and interpolated to the control volume surfaces. Convective and diffusive fluxes are interpolated using the second-order upwind differentiation scheme which uses the linear interpolation between two upwind centroids to approximate the fluxes at the surfaces of the control volume. Gradients are interpolated using the least squares method with Venkatakrishnan gradient limiters. The semi-implicit method for pressure-linked equations (SIMPLE) algorithm is used as the default segregated pressure-velocity coupling scheme. [star-ccm+] [what convection schemes for flow variables? and SIMPLE]

[what to mention later: volume fraction convection, semi-implicit treatment, blending schemes and stuff.]

1.3 Objectives and Outline

[———under construction —————]

- we describe the theoretical framework involved in modelling the two phenomenon
- we simulate the 1D stefan problem and 2D RT instability using the vof method. we also compare the emp framework in its ability to simulate the 2D RT instability?
- we simulate the infiltration of the evaporation chamber using the vof method, discuss the spatial and temporal convergence and a qualitative behaviour of the flow field.

The two main physical processes in the experiment - infiltration and evaporation - not only occur at distinct points in time but also require distinct mathematical treatment in their simulation. Thus, dividing the experiment into two subproblems -

- the infiltration of heated water in the evaporation chamber exposed to ambient temperature conditions,
- and the evaporation of superheated water in the evaporation subject to depressurisation -

this thesis investigates the feasibility of modelling the two processes in the commercial CFD software tool - Star-CCM+.

[to add: overview and the description of the structure]

2 Modelling the Physical Problem

This section introduces the mathematical framework and governing equations used to model the two-phase flow problems, laying the foundation for subsequent numerical simulations. The the rationale behind averaging techniques for multiphase flow modelling is explained and the macroscopic formulations used to represent complex two-phase dynamics introduced. The 'one-fluid' and 'two-fluid' models are detailed.

2.1 Evaporation: Phenomenological Overview and Literature Review

Evaporation processes are characterised based on whether they occur at saturation temperatures or not. Evaporation at saturation temperatures, known as vaporisation, is a much faster process in comparison to evaporation at normal [4]. In cases where vaporisation occurs along a heated wall, most CFD models distribute the energy transfer by separating the wall heat flux into two components: latent heat absorption associated with bubble vaporisation and sensible heat transferred to the liquid through conduction, convection, and quenching processes. The partitioning depends on the area fractions covered by liquid and vapour phases. Empirical results or models of subprocesses have established correlations between the heat flux and wall superheat.

The sub-processes that are primarily taken into account in such determinations are:

- number density of nucleation sites,
- diameter of bubble at time of detachment from the surface,
- and the frequency of bubble detachment.

Nucleation affects the number and distribution of bubbles in the fluid. Walls, particularly so in case they're heated, are more susceptible to be nucleation sites for bubble growth as the activation energy for bubble growth is much lower. This nucleation can be modelled through the domain as homogenous or heterogenous - where nucleation sites are modelled using a presumed PDF function or be determined by empirical correlations. [5]

After a stable bubble nucleus has been formed, the growth of the bubble becomes pertinent in the evaluation of the diameter at departure and the interphase energy and mass transfer. In the case of an ideal spherical bubble undergoing expansion in a uniformly superheated liquid, the rate of growth is determined by the surface tension, liquid inertia, and the difference in pressures inside and outside the bubble. Surface tension dominates and restricts the initial bubble growth. This initial rapid expansion of the bubble results in the formation of a thin liquid layer separating the vapour from the wall surface. This microlayer can be a significant contributor to the bubble growth and heat transfer process. As bubble radius increases, the liquid layer surrounding the bubble cools due to evaporation, resulting in a decrease in vapour pressure. The bubble growth dynamics are initially governed by thermal diffusion and inertia. Over time, inertial effects become less important as the bubble expands more slowly. Further growth is determined primarily by interphase energy exchange. [6]

2.1.1 Effects of low pressure and gravity conditions

The absence of significant gravitational forces alters the dynamics of vaporisation - affecting natural convection, buoyancy driven phenomenon, and bubble dynamics. Under terrestrial gravity, convection currents develop under thermal gradients, thus bringing cooler liquid to the heated surface as the vapour bubble rises due to buoyoancy. This density driven natural convection is affected in low gravity conditions, maintaining the thermal gradient. Bulk fluid movement is affected and heat and mass transfer processes shift from being convection dominated to being diffusion dominated. [7, 8]

Under low gravity conditions, vapour bubbles tend to remain near the heat source, growing significantly larger and coalescing with smaller bubbles. Prolonged bubble presence on the surface results in drying out of the surface and unsteady rise in surface temperatures.[9] [Mention some numbers] In conditions where the dominance of surface tension forces becomes more pronounced, the surface tension gradients can induce fluid circulation. This is known as Marangoni convection and is observed to affect bubble growth and heat transfer. [10]

Reduced system pressure impacts the overall intensity of heat transfer. The onset of nucleate boiling also requires higher superheat values in low pressure conditions. Decrease in pressure leads to a lower number of active nucleation sites and increases microlayer thickness found at the bubble-wall interface. The local heat flux density from microlayer evaporation is impacted and the microlayer evaporation area and depletion time is increased. The system is also susceptible to the formation of a continuous vapour film at the surface due to coalescing bubbles unable to escape the surface, reducing the overall heat transfer. [11]

The affected heat transfer and bubble growth phenomenon further result in lower critical heat fluxes and heat transfer efficiency. [12, 13]

2.1.2 Vaporisation in a superheated liquid

Flash evaporation is the intense vaporisation triggered through depressurisation of superheated fluids below the saturation pressure. This process induces a violent and transient heat and mass transfer, resulting in the explosive formation and growth of vapour bubbles throughout the liquid volume. The phenomenon is significant in desalination on ships, nuclear reactor safety, and venting of cryogenic propellant tanks, and the harvesting of low-temperature geothermal energy. [14, 15, 16, 17, 18, 19].

Early experiments that systematically studied flash evaporation in liquid pools and sprays in superheated water jets and early empirical correlations to determine the amount of generated vapour were established [20, 21, 22, 23]. For flash evaporation in static pools, the amount of liquid height increased the duration of the phenomenon and the rate of depressurisation increases the intensity. The final evaporated mass also increases proportional to the amount of superheat. While greater thermal inertia at larger depths slows the flashing process, the amount of total mass evaporated does not increase proportionally. [24, 14] It has been observed during flashing, that a liquid-vapour interface is generated on the free surface which then progresses into the superheated liquid [25, 26]. High-speed imaging and temperature measurements consistently show that flash evaporation proceeds through three distinct stages: bubble generation, boiling growth, and a boiling reduction stage [27]. For the final evaporated mass, under moderate superheat conditions $\Delta T \leq 35\,\mathrm{K}$, a strong linear scaling is observed [28, 16]. Recent experiments studying stationary volatile drop evaporation in microgravity report a 56% reduction in evaporation rate as compared to normal gravity conditions [29]. [need to add conclusion to this that ties into choice of phase change model later]

2.2 Modelling Multiphase Flows

The computational tribulations of directly solving local instantaneous governing equations throughout the entire domain necessitates dimensional reduction through macroscopic formulations. For two-phase flows (used interchangeably with multiphase flows), these formulations rely on averaging local flow field fluctuations while preserving continuum mechanics principles.

In the following, we use the subscript $i \in \{l, v\}$ to denote variables associate with the liquid (l) and vapour (v) phase respectively. Figure [FIG] displays a region Ω where two phases l, v, or equivalently p, s (primary and secondary), occupy regions $\Omega_l(t)$ and $\Omega_v(t)$ such that $\Omega = \Omega_l(t) \cup \Omega_v(t)$ is constant in time.

The boundaries of the domains - $\partial\Omega_l$, $\partial\Omega_v$, and $\partial\Omega$ - together define the interface that separates the two phases as

$$\mathcal{S}(t) \;=\; \Big(\partial\Omega_l \cap \partial\Omega_v\Big) \setminus \partial\Omega\,. \tag{2.1}$$

[Equation seems a bit pretentious or? also idk if is a good choice to make here for the notation] Eulerian averaging is related to the Eulerian description of the flow - the local instantaneous variables are averaged with the time and space variables being kept constant. Volume averaging admits a general representative volume element V bounded by ∂V, and time averaging uses a time interval at every spatial reference point $(\underline{\mathbf{x}}, t)$ in the region. The averaging volume V is partitioned into V_i for phase $i \in \{l, v\}$ with phase volume fraction

$$\alpha_i(\underline{\mathbf{x}}, t) \;=\; \frac{V_i}{V}\,, \qquad \alpha_l + \alpha_v = 1\,. \tag{2.2}$$

All averaging procedures generally utilise a phase indicator function that marks the presence of the phase in the averaging domain. Denoted as γ_i, the phase indicator function can be used to define the average of the general scalar or vector quantity ψ over the averaging volume V,

$$\overline{\psi_i}^{V} \;=\; \frac{1}{V}\int_V \gamma_i\,\psi_i(\underline{\mathbf{x}}, t)\,\mathrm{d}V\,, \tag{2.3}$$

and the phase-intrinsic average then is

$$\overline{\psi_i}^{i} \;=\; \frac{1}{V_i}\int_{V_i} \psi(\underline{\mathbf{x}}, t)\,\mathrm{d}V \;=\; \frac{\overline{\psi_i}^{V}}{\alpha_i}\,. \tag{2.4}$$

Substituting instantaneous variables with their averaged equivalents acts as a low-pass filter, smoothing out discontinuities at the phase interface. This process eliminates direct knowledge of interface topology, thus necessitating closure models and explicit interfacial transfer terms. Notably, while Star-CCM+ employs Euler-averaged equations for its multiphase models, the software documentation does not specify which averaging approach - time, volume, or a combination - is used. A thorough treatise on the averaging procedures can be found [here and here]. The development of the two-fluid model and one-fluid model for two-phase flows is detailed [here and here] respectively. [language] [1, 30, 31, 32] [drew, star-ccm+]

Formulations of the macroscopic field equations for multiphase flows can either treat each phase separately or consider the mixture as a whole. When the dynamics of the two phases are closely coupled, the two sets of governing equations can be summed up together which then along with the relevant interfacial transfer terms describe the behaviour of the mixture. In the two-fluid formulation, the set of the averaged governing equations is solved for each phase. The phases can evolve independently and are coupled through interfacial transfer terms. In both formulations, the transfer terms coupling the phases must also account for the relevant jump conditions. [1, 30, 33, 2]

Often a 'one-fluid' model is utilised to model separate flows due to its ease of implementation and computational efficiency. The phases are treated as one fluid with variable material properties that have an abrupt change at the interface. Interface capturing methods such as the volume-of-fluid (VOF) employ a marker function that distinguishes the phases and is advected with the evolution of the flow. This marker function can be used to reconstruct the interface. The governing equations are analogous to the mixture formulation of two-phase flows with no relative velocity between the phases. [34, 33, 31] [star-ccm+]

The following discussion presumes that the averaging procedure has been performed and discusses the governing equations in their averaged macroscopic form as solved in Star-CCM+. The quantities are assumed to be the phase intrinsic averages. For both the formulations, we assume the validity of the continuum hypothesis, that the interfaces are infinitely thin, and neglect intermolecular forces.

2.2.1 The One-Fluid Formulation

In order to account for the varying material properties when representing two-phase flow as a single fluid, the mixture properties are defined in the volume element V. The mixture density ρ, viscosity μ, and thermal conductivity k are defined as

$$\rho = \sum_i \alpha_i \rho_i \,, \tag{2.5a}$$
$$\mu = \sum_i \alpha_i \mu_i \tag{2.5b}$$
$$k = \sum_i \alpha_i k_i \,. \tag{2.5c}$$

Here, the properties ρ_i, μ_i, and k_i are phase intrinsic averages. Using the defined mixture density ρ, the velocity, specific heat, and enthalpy can be also defined as mass averaged mixture values.

$$\underline{\mathbf{v}} = \sum_i \alpha_i \rho_i \underline{\mathbf{v}}_i / \rho, \tag{2.6a}$$
$$C_p = \sum_i \alpha_i C_{p,i} \rho_i / \rho, \tag{2.6b}$$
$$H = \sum_i \alpha_i \rho_i \mathrm{H}_i / \rho + |\underline{\mathbf{v}}|^2/2 \,, \text{where } \underline{\mathbf{v}} \text{ is the mixture velocity.} \tag{2.6c}$$

Sometimes a relative velocity between the phases is considered, the diffusion velocity $\underline{\mathbf{v}}_{d,i}$ of phase i can be defined as

$$\underline{\mathbf{v}}_{d,i} = \underline{\mathbf{v}}_i - \underline{\mathbf{v}} \,. \tag{2.7}$$

The stress tensor $\underline{\underline{\mathbb{T}}}$ can be extended to include both the molecular and turbulent stresses. The various forces are accounted through the pressure p, gravitational acceleration $\underline{\mathbf{g}}$, and the body force vector $\underline{\mathbf{f}_\mathbf{b}}$. $\underline{\underline{\mathbb{I}}}$ is the second order identity tensor. The total energy of the system is given as E and the heat flux vector is denoted by $\underline{\dot{\mathbf{q}}''}$. Given mass, momentum, and energy sources for the phase i as S_i^α, $\underline{\mathbf{S}}_i^\alpha$, and S_i^E respectively, the governing equations for the one-fluid formulation are given as follows:

$$\frac{\partial}{\partial t}\int_V \rho \,\mathrm{d}V + \int_{\partial V} \rho \underline{\mathbf{v}} \cdot \mathrm{d}\underline{\mathbf{a}} = \int_V \sum_i \mathrm{S}_i^\alpha \rho_i \,\mathrm{d}V \,, \tag{2.8a}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \rho \underline{\mathbf{v}} \,\mathrm{d}V + \int_{\partial V} \rho \underline{\mathbf{v}} \otimes \underline{\mathbf{v}} \cdot \mathrm{d}\underline{\mathbf{a}} &= \int_{\partial V} \left(-p\underline{\underline{\mathbb{I}}} + \underline{\underline{\mathbb{T}}}\right) \cdot \mathrm{d}\underline{\mathbf{a}} + \int_V \rho \underline{\mathbf{g}} \,\mathrm{d}V + \int_V \underline{\mathbf{f}}_b \,\mathrm{d}V \\ &\quad - \sum_i \int_{\partial V} \alpha_i \rho_i \underline{\mathbf{v}}_{d,i} \otimes \underline{\mathbf{v}}_{d,i} \cdot \mathrm{d}\underline{\mathbf{a}} + \int_V \sum_i \underline{\mathbf{S}}_i^\alpha \,\mathrm{d}V\end{aligned} \tag{2.8b}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \rho \mathrm{E} \,\mathrm{d}V + \int_{\partial V} \left[\rho H \underline{\mathbf{v}} + \sum_i \alpha_i \rho_i \mathrm{H}_i \underline{\mathbf{v}}_{d,i}\right] \cdot \mathrm{d}\underline{\mathbf{a}} &= -\int_{\partial V} \underline{\dot{\mathbf{q}}''} \cdot \mathrm{d}\underline{\mathbf{a}} + \int_{\partial V} \underline{\underline{\mathbb{T}}} \cdot \underline{\mathbf{v}} \,\mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{f}}_b \cdot \underline{\mathbf{v}} \,\mathrm{d}V \\ &\quad + \int_V \sum_i S_i^E \,\mathrm{d}V\end{aligned} \tag{2.8c}$$

The computational efficiency and ease of implementation of the one-fluid formulation is limited by increased coupling of the phases and complex interfacial dynamics - as the averaging volume must be able to capture the various scales involved in the interphase interactions. As jump conditions are not explicitly enforced, interfacial terms must be added to the governing equations at the interface which brings in numerical complications. Many existing one-fluid formulations in the literature are analytically inconsistent under vaporization. Under conditions of phase change, the conservative and non-conservative forms of the governing equations for various formulations are found to not be equivalent and that the momentum and energy jump conditions are not recovered. [35]. Phase change is generally modelled using source terms in the volume fraction equation and the energy equation [36, 33].

2.2.2 The Two-Fluid Formulation

Considering the volume element V as described in the previous section, the volume fractions of the two phases α_i $i \in \{l, v\}$ must satisfy the continuity constraint

$$\sum_i \alpha_i \;=\; 1\,. \tag{2.9}$$

The volume of the phase i is given as

$$V_i = \int_V \alpha_i \,\mathrm{d}V\,, \tag{2.10}$$

The ratio V_i/V of the effective volume of the phase i to the total volume is defined as the void fraction of the phase.

The averaged macroscopic equation for the conservation of mass is given in (2.12). The density and velocity of the phase i is denoted by ρ_i and $\underline{\mathbf{v}}_i$. The interfacial mass transfer unit volume between the two phases is denoted by Γ and the surface area vector of the volume element is denoted by $\underline{\mathbf{a}}$. With an additional phase mass source term $\mathcal{S}_i^\alpha$, the conservation of mass for a generic phase i is

$$\frac{\partial}{\partial t}\int_V \alpha_i\rho_i \,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\underline{\mathbf{v}}_i\cdot \mathrm{d}\underline{\mathbf{a}} \;=\; \int_V \Gamma_i \,\mathrm{d}V + \int_V \mathrm{S}_i^\alpha \,\mathrm{d}V\,. \tag{2.11}$$

The interfacial mass transfer term Γ_i represents the rate of mass transfer from the phase i. The interaction area density a_{lv} specifies the interfacial area available in the volume element over which the mass transfer occurs.

$$\Gamma_i \;=\; a_{lv}\,\dot{m}_i \;=\; \rho_i\underline{\mathbf{n}}_i\cdot\left(\underline{\mathbf{v}}_i - \underline{\mathbf{v}}_{\mathcal{S}}\right). \tag{2.12}$$

The term $\rho_i\underline{\mathbf{n}}_i\cdot\left(\underline{\mathbf{v}}_i - \underline{\mathbf{v}}_{\mathcal{S}}\right)$ consists of the local instantaneous variables instead of the averaged quantities. The interface normal vector and interface velocity are given as $\underline{\mathbf{n}}_i$ and $\underline{\mathbf{v}}_{\mathcal{S}}$ respectively.

The governing equation for the conservation of momentum is given in (2.14). The equation describes the change in momentum of the phase i due to the effects of the fundamental forces in the system along with the effects of any internal forces or potentials $(\underline{\mathbf{F}_{\mathbf{int}}})_i$ or any phase momentum source terms $\underline{\mathbf{S}}_i^\alpha$. The pressure and gravitational acceleration is given as p and $\underline{\mathbf{g}}$ respectively. The stress tensor $\underline{\underline{\mathbb{T}}}_i$ comprises of the molecular and turbulent stress tensors. Turbulence in addressed in 2.3.

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\underline{\mathbf{v}}_i \,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\underline{\mathbf{v}}_i\otimes\underline{\mathbf{v}}_i\cdot\,\mathrm{d}\underline{\mathbf{a}} \;=\; &\int_V \left(-\alpha_i\nabla p + \alpha_i\rho_i\underline{\mathbf{g}} + (\underline{\mathbf{F}_{\mathbf{int}}})_i\right)\,\mathrm{d}V + \int_{\partial V}\left[\alpha_i\left(\underline{\underline{\mathbb{T}}}_i\right)\right]\cdot\,\mathrm{d}\underline{\mathbf{a}}\\ &+\int_V \underline{\mathbf{S}}_i^\alpha\,\mathrm{d}V + \int_V \underline{\mathbf{M}}_i\,\mathrm{d}V\,.\end{aligned} \tag{2.13}$$

The momentum transfer term $\underline{\mathbf{M}}_i$ accounts for the momentum transfer from the phase i to the interface. This term also accounts for the momentum transfer due to interfacial mass transfer and is further detailed in Section 3.4. It is only non-zero in case of a relative motion between the phases and is subject to the constraint

$$\underline{\mathbf{M}}_i \;=\; a_{lv}\left(\left(-p_i + \underline{\underline{\mathbb{T}}}_i\right)\cdot\underline{\mathbf{n}}_i - \dot{m}_i\underline{\mathbf{v}}_i\right)\,. \tag{2.14}$$

In case of constant surface tension, the dynamic condition that the effects of the hydrodynamic forces are in equilibrium with the forces due to surface tension. Denoting the part of the interface $\mathcal{S}$ in the volume element V as $\mathcal{S}\cap V$, the dynamic jump condition can be written as

$$\underline{\mathbf{M}}_l + \underline{\mathbf{M}}_v \;=\; \frac{1}{V}\int_{\mathcal{S}\cap V} \underline{\mathbf{f}}_\sigma\,\mathrm{d}V\,. \tag{2.15}$$

The averaged macroscopic formulation of the conservation of energy in the two-fluid formulation is shown in (2.17). The equation accounts for the change of the total energy E_i in the volume V due to the work

done by line, surface, and volume (body) forces on the volume element, any heat transfer (in this specific case, only conduction and convection are considered), and due to the interfacial mass transfer in the volume element.

$$\frac{\partial}{\partial t}\int_V \alpha_i \rho_i E_i \, \mathrm{d}V + \int_{\partial V} \alpha_i \rho_i H_i \underline{\mathbf{v}}_i \cdot \mathrm{d}\underline{\mathbf{a}} = \int_{\partial V} \alpha_i (k_{\mathrm{eff}})_i \nabla T_i \cdot \mathrm{d}\underline{\mathbf{a}} + \int_{\partial V} \underline{\underline{\mathbb{T}}}_i \underline{\mathbf{v}}_i \cdot \mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{f}_{\mathbf{b}}}_i \cdot \underline{\mathbf{v}}_i \, \mathrm{d}V + \int_V \mathrm{S}_{u,i} \, \mathrm{d}V + \Lambda. \tag{2.16}$$

Having each transfer process of each phase balanced separately, the two-fluid formulation is much more complex and requires closure models for each interaction. However, the two-fluid formulation is a much more complete and general description of two-phase flow - accounting for the dynamic and no-equilibrium interactions between the phases. The mixture or the two-fluid model do not retain information about the interface topology. Given a certain void fraction of a phase the two models are unable to comment on the nature of the flow regime. [1].

2.3 Modelling Turbulence

Flows are generally turbulent. Turbulence is characterised by irregular fluid motion dominated by eddies and vortices across a wide range of spatial and temporal scales. The largest eddies extract kinetic energy from the mean flow, which is then, through a cascade, transferred to the eddies at the smallest scale where the energy is dissipated due to viscosity, raising the internal energy of the fluid [37, 38].

RANS models result from decomposing flow variables into mean and fluctuating components, $\phi = \overline{\phi} + \phi'$, applying Reynolds averaging, which filters out turbulent fluctuations below the averaging scale [39, 40]. This operation yields additional Reynolds stress terms representing momentum transport by turbulence, which require closure. In order to draw closure relations, eddy-viscosity models of turbulence assume that:

- the turbulent scalar flux vector aligns with the mean scalar gradient.
- the Reynolds stress tensor is proportional to the mean velocity strain rate through an eddy viscosity.

While these assumptions simplify modelling, they tend to be, more often than not, invalid as they do not fully capture turbulence physics, especiialy in anisotropic or non-equilibrium flow [41]. This limitation can be significant in regions with complex strain rates or strong secondary flows.

Although turbulence can be modelled in various ways to various levels of resolutions, turbulence in this study has been modelled using the Shear Stress Transport (SST) k-ω model. The SST k-ω model is a two-equation eddy-viscosity turbulence model based on the Reynolds-Averaged Navier-Stokes (RANS) equations. The model combines the advantages of the k-ω model near walls - which accurately resolves boundary layer behaviour - and the k-ε model in the free stream, enhancing robustness and accuracy in separated flows [42]. The SST approach uses a blending function to smoothly transition between these models, which is especially effective in flows with strong adverse pressure gradients and separation, such as multiphase cavity flows [43]. Another blending function limits the eddy viscosity in wake regions and a cross-diffusion term ensures consistency between the k-ω and k-ϵ models.

Since the form of the RANS governing equations remains similar to the governing equations described in the previous section, they are not explicitly written here but can be found in the Star-CCM+ documentation [starccm]. The additional equations accounting for the transport of the turbulent kinetic energy and dissipation rate is given in 2.18

$$\frac{\partial \rho k}{\partial t} + \nabla(\rho k \underline{\bar{\mathbf{v}}}) = \nabla\left[(\mu + \sigma_k \mu_t)\nabla k\right] + P_k - \rho\beta^* f_{\beta^*}(\omega k - \omega_0 k_0) + S_k, \tag{2.17a}$$

$$\frac{\partial \rho \omega}{\partial t} \nabla(\rho \omega \underline{\bar{\mathbf{v}}}) = \nabla\left[(\mu + \sigma_\omega \mu_t)\nabla \omega\right] + P_\omega - \rho\beta f_\beta(\omega^2 - \omega_0^2) + S_\omega, \tag{2.17b}$$

The turbulent viscosity, μ_t, is calculated as $\mu_t = \rho k T t$, where T_t is the turbulent time scale. where σ_k and σ_ω are the Prandtl numbers for k and ω respectively.

2.4 Modelling Subproblem 1

Not considering the mixing effects that might occur at later times, the infiltrating water into the evaporation chamber is a separated flow problem due to the sharp interface separating the two phases - water and air.

The objective - we want to understand the development of the flow field, understand the amount of sloshing that can be expected in the system, and the amount of mixing that happens during the injection process and what the result flow system might look like before the evaporation step.

The fundamental forces active in the system are the inertial, gravitational, viscous, pressure, and surface tension forces. The governing equations are analogous to the governing equations of the homogenous two-fluid flow model[1]. The flow field is described using the primary variables for each phase - ρ_i, $\vec{u}_i$, E_i.

- what we're looking for and why we're interested in this
- phases involved, forces active and shit
- assumptions we make
- variables we care about
- governing eqs (?)
- 3D vs 2D

2.5 Modelling Subproblem 2

- from the top

[1] two-fluid formulation with the phase velocities assumed to be constant

3 Nums

Star-CCM+ uses the one-fluid formulation to implement the volume-of-fluid method. The two-fluid model is implemented as the eulerian multiphase (EMP) model. The eulerian multiphase model is further equipped to resolve large scale interfaces and thus can model both disperse and separated flow regimes. [34] [star-ccm] This chapter details the numerical implementation of both the models in Star-CCM+ and discusses the closures relations used to close the system of equations.

3.1 VOF

The volume-of-fluid (VOF) method is an interface capturing method that utilises the governing equations of the one-fluid model as detailed in Section 2.2.1. The method presumes that the computational mesh is fine enough to resolve the interface topology. Using the volume fraction as the marker function - the interface is assumed to be present in cells with volume fraction values between 0 and 1. The distribution of the phase i is driven by the volume fraction transport equation.

$$\frac{\partial}{\partial t}\int_V \alpha_i \,\mathrm{d}V + \int_{\partial V} \alpha_i \underline{\mathbf{v}}_i \cdot \mathrm{d}\underline{\mathbf{a}} = \int_V \left(S_i^\alpha - \frac{\alpha_i}{\rho_i}\frac{D\rho_i}{Dt} \right) \mathrm{d}V - \int_V \frac{1}{\rho_i}\nabla(\alpha_i \rho_i \underline{\mathbf{v}}_{d,i}) \,\mathrm{d}V \,. \tag{3.1}$$

In case of two phases, the volume fraction transport is solved only for the first phase, thus in each volume element, the volume fraction of phase j is given as $\alpha_j = 1 - \alpha_i$.

The High Resolution Interface Capturing (HRIC) scheme is the default volume fraction convection scheme used. The scheme is based on the normalised variable diagrams. The scheme blends the downwind scheme with the upwind scheme to resolve the interface parallel or perpendicular to the flow direction. The calculated normalised face values of the volume fractions are then corrected in order to conform to the local Courant number. The spurious oscillations that may arise in the volume fraction field due to meshes with large aspect ratios are suppressed by the artificially smoothening the volume fraction gradient across the interface. [star-ccm+]

3.2 EMP

Star-CCM+ implements the governing equations of the two-fluid model as detailed in Section **??** as the eulerian multiphase (EMP) model. The two-fluid model is predominantly used to simulate dispersed flows and the mesh resolution is assumed to be coarse enough to not capture the interface topology between the phases. The EMP model, however, utilises the multiple flow regime framework to allow the simulation of multiscale flows. Based on increasing volume fraction of the secondary phase, α_s, in a finite volume cell, the topology transitions through three principal regimes [34] [starccm], where the secondary phase is:

- dispersed in the primary phase (first dispersed regime, $0 \leq \alpha_s \leq \alpha_{fr}$)
- separated from the primary phase through a large scale interface (interfacial flow regime, $\alpha_{fr} \leq \alpha_s \leq \alpha_{sr}$)
- continuous while having the primary phase dispersed (second dispersed regime, $\alpha_{sr} \leq \alpha_s \leq 1$)

The first and second regime terminuses, α_{fr} and α_{sr}, mark the values of α_s that defines the end of the first dispersed and large scale interface regimes respectively. Star-CCM+ uses a regime-specific closure models for the interfacial momentum and energy terms. These models are blended using a weighting scheme according to the calculated proportion of each regime in the cell.

A variety of convection schemes for the transport of volume fraction are available. The Adaptive Interface Sharpening (ADIS) scheme can be used in the multiple flow regime framework to avoid spurious oscil-

lations near sharp interfaces due to the upwind schemes, or artificial sharpening of otherwise smooth regions due to HRIC. ADIS uses the HRIC scheme in the vicinity of a large scale interface and uses a standard upwind scheme away from the sharp interface. [starccm]

3.3 Star-CCM+ Stuff - needs a better heading

[relevance?] The implementation of the SST k-ω model in Star-CCM+ can be further controlled to adjust the realisability. Realisability is the positive semi-definiteness of the reynolds stress tensor. The specification of turbulence is done using the turbulence intensity and the length scale. The inherent anisotropy in turbulence is accounted for with quadratic and cubic constitutive options in the modelling of reynolds stresses. The parameters in the model are left on their default values as they are empirically modelled.

At large scale interfaces, the near-interface flow often exhibits laminar characteristics with the development of boundary layers. However, in situations with significant velocity gradients, turbulent fluctuations can arise in the regions adjacent to the interface (especially when the interface is mobile or highly distorted). Recognizing that standard RANS turbulence models tend to overpredict turbulence intensity near phase boundaries, STAR-CCM+ allows the user to apply turbulence damping strategies (e.g., vorticity limiters) which suppress excessive eddy viscosity and restore physically consistent laminar sublayers adjacent to the interface.

3.4 Momentum Closures

Closures are critical to the accurate prediction of the relevant flow variables and the distribution of volume fraction [44]. Existing literature reviews the overwhelming number of closures available that yield a range of varyinh performance along with their comparison with experimental data, however literature presenting CFD simulations tend to choose 'result-oriented' closures that aren't transferable to a wider range of flow situations [45, 46]. Literature admits that the industry relies on the tunable parameters in the closures to match the specific experimental data being simulated or the arbitrariness of this choice to get improved flow predictions [[20] rzehak, kriebitzsch; silva [21] (sugrue)].

Models may also provide satisfying predictions for certain parameters or situations while failing in others and a general inconsenus prevails while discussing the effects of interfacial forces. Early work has tended to neglect the effects of lift or virtual mass forces, or chosen to use constant coefficients to describe them while using formulations for the drag forces based on relevant non-dimensional numbers. [47] The effect of non-drag forces, however, tends to be as significant on the distribution of phases. [liao2014-flashboiling] Since it is difficult to separate the effects of the various forces in actual experiments, the development of these closures assumes certain idealisations when they're not partly or wholly based on empirical data, rendering them highly flow dependent [troshko and hassan, [15] in Sugrue]. The authors of [troshko and hassan] critiques the methodology of resolving a complex interfacial momentum transfer through a linear superposition of several simple forces as the basis of the discrepancy.

The industry has, however, attempted to establish generality that can be transferred to multiple flow situations and has proposed baseline frameworks. There is a general consensus that the role of turbulence is underestimated [48]; current models also aim to reduce uncertainty and inconsistency in model selection [48], adopt closures that avoid "case-specific adjustments to unphysical coefficients and tunable parameters" [44], and utilise more physics based closures that are more physically consistent [47, 48].

Drag force is generally more dominant than other forces as they tend to be a magnitude higher than other forces [49]. Most drag models perform equally well at low gas volume fractions. However as they are based on ideal conditions, they do not account for bubble swarm effects at higher gas volume fractions and show reduced accuracy. Certain swarm corrections do exist, but they are averaged along a radial profile when the swarm effects are also felt in the bulk. [44, 46, 48] The accurate modelling of interaction area density

and length becomes relevant for the calculation of the drag force. Using the linearised drag coefficient, A_D, and the relative velocity between the phases $\underline{\mathbf{v}}_r = \underline{\mathbf{v}}_j - \underline{\mathbf{v}}_i$, Star-CCM+ implements the drag force as,

$$\underline{\mathbf{F}}_{ij}^D = A_D \underline{\mathbf{v}}_r \tag{3.2}$$

The linearised drag coefficient is related to the standard definitions of the drag coefficient C_D using

$$A_D = C_D \frac{1}{2} \rho_c |\underline{\mathbf{v}}_r| \frac{a_{cd}}{4} \tag{3.3}$$

where a_{cd} is the interfacial area density and $a_{cd}/4$ represents the projected area of the equivalent spherical particle.

The lift force models are particularly sensitive and uncertain in their performance [46]. They suffer from the same discrepancy between performance in low and high gas volume fractions. Moreover, existing lift force implementations in Star-CCM+ fail to replicate the experimental data they were derived from. [44] Assuming α_p is the volume fraction of the primary (continuous) phase, Star-CCM+ implements the lift force as

$$F_{ij}^L = C_{L,\text{eff}} \alpha_d \rho_p \left[\underline{\mathbf{v}}_r \times \left(\nabla \times \underline{\mathbf{v}}_p \right) \right]. \tag{3.4}$$

A turbulent dispersion force is supposed to act on the gas phase and spreads this dispersed phase due to turbulence in the continuous phase. This is typically modelled as as a diffusion term proportional to the turbulent kinetic energy gradients [28]. The force is theorised to arise from the averaging of the fluctuations of the bubbles due to the turbulence.

$$F_{ij}^{TD} = A_D \underline{\mathbf{v}}_{TD} = A_D \underbrace{C_0 \frac{\nu_c^t}{\sigma_\alpha} \underline{\underline{\mathbb{I}}}}_{\underline{\underline{\mathbb{D}}}_{TD}} \cdot \left[\nabla \ln(\alpha_d) - \nabla \ln(\alpha_c) \right] \tag{3.5}$$

The closure relation by Burns [burns] uses the Favre averaging of the drag force and can be expressed as

$$F_{ij}^{TD} = C_D C_{TD} \frac{\mu_l^t}{\rho_c \sigma_{TD}} \underline{\mathbf{v}}_{TD}, \tag{3.6}$$

where C_D is the drag coefficient, C_{TD} is an adjustable turbulent dispersion calibration coefficient, μ_l^t is the liquid turbulent viscosity, and σ_{TD} is the turbulent Prandtl number.

Most lift coefficient correlations don't account for the decrease of the magnitude of the lift force to zero near the walls. This low gas volume fraction at the walls has historically been explained through a liquid film forming at the walls repelling the bubbles. This has been demonstrated to not actually occur in practice. The wall lubrication force has thus been criticised as lacking any physical justification. [44, 48] Newer closures [lubchenko] utilise a turbulent dispersion force based formualtion to maintain the desired gas volume fraction and use the closure instead as a correction term to resolve the phase distribution at the boundary lost due to the averaging process. Star-CCM+'s implementation of the wall lubrication force

$$F_{ij}^{WL} = -C_{WL}(y_w) \alpha_s \rho_p \frac{|\underline{\mathbf{v}}_r - (\underline{\mathbf{v}} \cdot \underline{\mathbf{n}}) \underline{\mathbf{n}}|}{d_b} \underline{\mathbf{n}} \tag{3.7}$$

can be changed to implement alternative models instead by defining user defined field functions for C_{WL}.

Surface tension, although an interfacial force, is implemented using the Continuum Surface Force approach as a volumetric force in the VOF formulation. Divided into normal and tangential components and using the mean curvature of the free surface $\kappa = -\nabla \cdot (\nabla \alpha_i / |\nabla \alpha_i|)$, the volumetric surface tension force is given as

$$\underline{\mathbf{f}}_\sigma = \underbrace{-\sigma \nabla \cdot \frac{\nabla \alpha_i}{|\nabla \alpha_i|} \nabla \alpha_i}_{\underline{\mathbf{f}}_{\sigma,n}} + \underbrace{(\nabla \sigma)_t |\nabla \alpha_i|}_{\underline{\mathbf{f}}_{\sigma,t}} \tag{3.8}$$

The EMP treats surface tension by splitting it among the phases occupying the cell using the local volume fractions. The pressure gradient is then the summation of the individual phase surface tension contributions $\underline{\mathbf{f}}_{\sigma,i}$, [34] [starccm],

$$\underline{\mathbf{f}}_{\sigma,i} = \alpha_i \sigma \kappa \nabla \alpha_i. \tag{3.9}$$

3.5 Closures: Under Construction

- Interphase mass transfer (evaporation/condensation) Represents phase change, modeled as interfacial flux determined by local thermodynamic properties (temperature, concentration, pressure—often using kinetic models for flash evaporation).
- Interphase energy transfer Encompasses latent heat and sensible heat exchange at the interface and throughout the phases (essential for accurate phase change modeling).
- Nucleation (wall and bulk)
- Interfacial area changes due to bubble growth, coalescence, breakup
- Local pressure and temperature gradients driving simultaneous mass and energy transfer

3.6 Vague and Miscellaneous

4 Computational Methodology and Results

This chapter covers the simulation setup of liquid infiltration of the evaporation chamber. The objective is to simulate the water injection and extract quantities of interest to better estimate the initial conditions for the second task. The physics models used in the simulation of the problem are validated against the 1D Stefan problem and the 2D Rayleigh-Taylor instability. The problems are introduced and the computational methodology and result of these simulations is shortly discussed. The chapter concludes with the description of the computational methodology, results, and comments on the 2D simulation of the infiltration problem.

4.1 Stefan Problem

Phase change processes are characterized by the movement of an interface separating two distinct phases, where one phase expands as another contracts due to thermal or non-thermal mechanisms [50]. The mathematical model governing such moving boundary problems is known as the Stefan system [50], and the analysis of such a system forms the basis of the Stefan problem.

In the classical Stefan problem, a fixed-temperature boundary propagates a melting or freezing front into the domain, with interface movement determined by the balance of latent and conducted heat. [51] With uniform initial temperature, constant boundary temperature and thermophysical properties in each phase, the semi-infinite 1D Stefan problem is the simplest, explicitly solvable problem of phase change. As the problem admits an analytical solution and reduces to the standard transient heat conduction problem when latent heat is neglected, it serves as a rigorous benchmark for validating phase-change modelling and the correct implementation of the energy equation. [51, 52, 53].

Assuming that the region in space, $\Omega \subset \mathbb{R}^2$, occupied by the material is the 2D semi-infinite rectangle $\Omega = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \in [0, \infty), y \in [0, \Delta h]\}$, with the solid phase and the liquid phase occupying the regions Ω_s and Ω_l respectively such that $\Omega = \Omega_s \cup \Omega_l$. At any given time, t, the two phases are separated by the interface, $X(t)$, that marks the change in phase of the material. The boundary of the domain Ω is denoted by $\partial\Omega$.

The problem space, $\mathcal{D} = \{[t_0, t_f] \times \Omega\}$, is the spatial and temporal domain occupied by the material. With the heat equation governing the temperature field in the liquid region, the Stefan condition at the interface, and the respective boundary conditions, the system of equations defining the one-dimensional Stefan problem can be written as,

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \text{ for } 0 < x < X(t), t_0 < t < t_f \tag{4.1a}$$

$$-k \frac{\partial T(X(t), t)}{\partial x}\Big|_{X^-(t)} = \rho L \frac{\partial X(t)}{\partial t}, \text{ for } t_0 < t < t_f \tag{4.1b}$$

$$T(x, 0) = T_m, \text{ on } \Omega \tag{4.1c}$$

$$T(0, t) = T_l, \text{ on } \partial\Omega(0, t) \text{ for } t_0 < t < t_f \tag{4.1d}$$

$$\nabla T \cdot \underline{\mathbf{n}} = 0, \text{ on } \partial\Omega - \partial\Omega(0, t) \tag{4.1e}$$

An analytical solution to the system of equations is given as

$$\hat{T}(x, t) = T_l - (T_l - T_m) \frac{\text{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right)}{\text{erf}\left(\frac{X(t)}{2\sqrt{\alpha t}}\right)}, \tag{4.2}$$

$$\hat{X}(t) = 2\lambda\sqrt{\alpha t}, \tag{4.3}$$

where λ is the unique root of the monotonic function

$$f(\lambda) := \frac{C_p(T_m - T_l)}{L} - \frac{\lambda e^{-\lambda^2}}{\sqrt{\pi}\mathrm{erf}(\lambda)}$$

The error function $\mathrm{erf}(x)$ is defined as $\mathrm{erf}(x) := 2/\sqrt{\pi}\int_0^x e^{-y^2}\,dy$. The physical parameters and computational methodology are summarized in Table 1

The simulations assume a laminar but unsteady flow. The computational domain of the simulation is chosen to be $[0, L] \times [0, 4\Delta h]$, where Δh is the cell width of the discretised domain. The domain is discretised using a trimmed cell mesher [CONFIRM] with uniform quadrilateral elements. The volume of fluid method with first order HRIC is used to reconstruct the interface. The convection and diffusion terms are discretised using the second-order upwind scheme and the least squares method was used to interpolate the values of the gradients from the cell centres onto the face centres. The transient term is discretised using the implicit Euler-method which is first-order accurate. All four boundaries are assigned a no-slip shear stress specification. The boundary at $x = 0$ is assigned a constant temperature, $T_l = 300\,\mathrm{K}$, while the other boundaries are assumed to be adiabatic. Since Star-CCM+ sets under- and over-relaxation factors conservatively, the values are left unchanged.

Simulation Setup	
Domain	2D rectangular domain
Mesh size, Δh	2.0×10^{-3} m
Time discretization	Implicit unsteady, 1st order Euler
Time step, Δt	0.5 s
Physical Properties	
Density, ρ	$1000.0\,\mathrm{kg{\cdot}m^{-3}}$
Dynamic viscosity, μ	8.8871×10^{-4} Pa·s
Specific heat, C_p	$4200.0\,\mathrm{J{\cdot}(kg{\cdot}K)^{-1}}$
Latent heat of fusion, L	$1.0 \times 10^5\,\mathrm{J{\cdot}kg^{-1}}$
Thermal conductivity, k_l	$2.33\,\mathrm{W{\cdot}(m{\cdot}K)^{-1}}$
Thermal conductivity, k_s	$0.62\,\mathrm{W{\cdot}(m{\cdot}K)^{-1}}$
Melting temperature, T_m	273 K
Numerical Methods	
Multiphase model	VOF with HRIC convection
Flow model	Laminar
Spatial discretization	2nd order (energy), 1st order (HRIC)

Table 1: CFD simulation parameters for Stefan melting-solidification problem

The physical parameters and the computational methods involved in the simulation setup are summarised in Table 1. The evolution of the temperature field for the simulation with cell size, $\Delta h = 5.0 \times 10^{-3}$ m is shown in Figure 1. The interface is observed to move towards the right with increasing simulation time. The temperature of the system can also be seen to decrease from the Dirichlet boundary at $x = 0$ towards the melting temperature, T_m, at the interface, after which it stays constant.

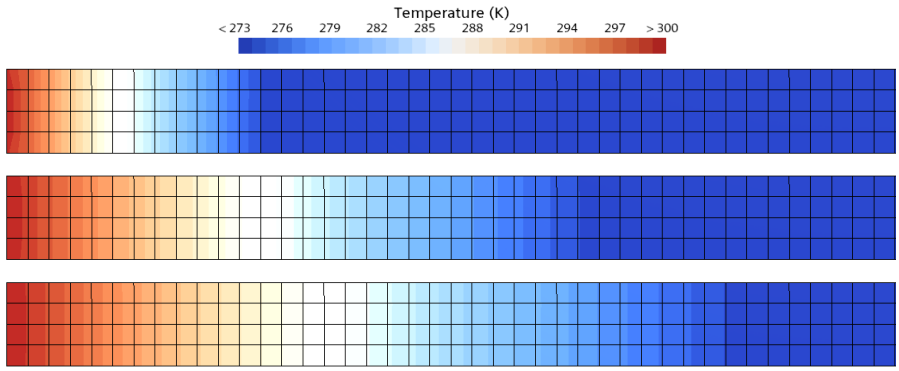


Figure 1: Evolution of the phase change interface

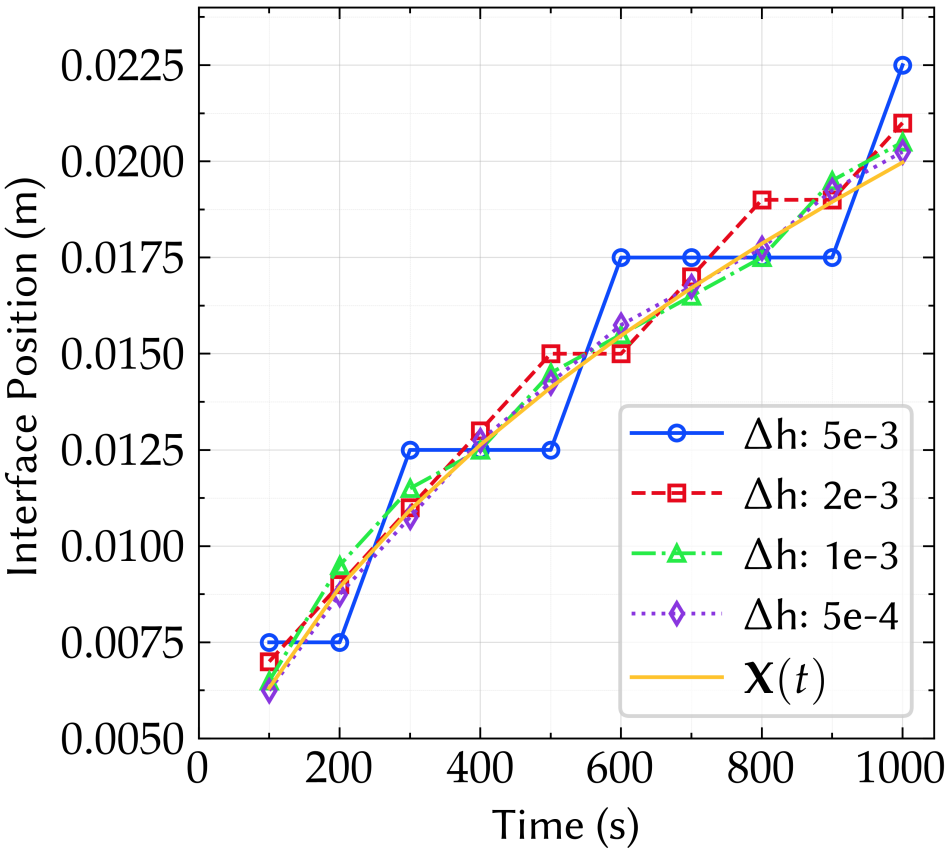
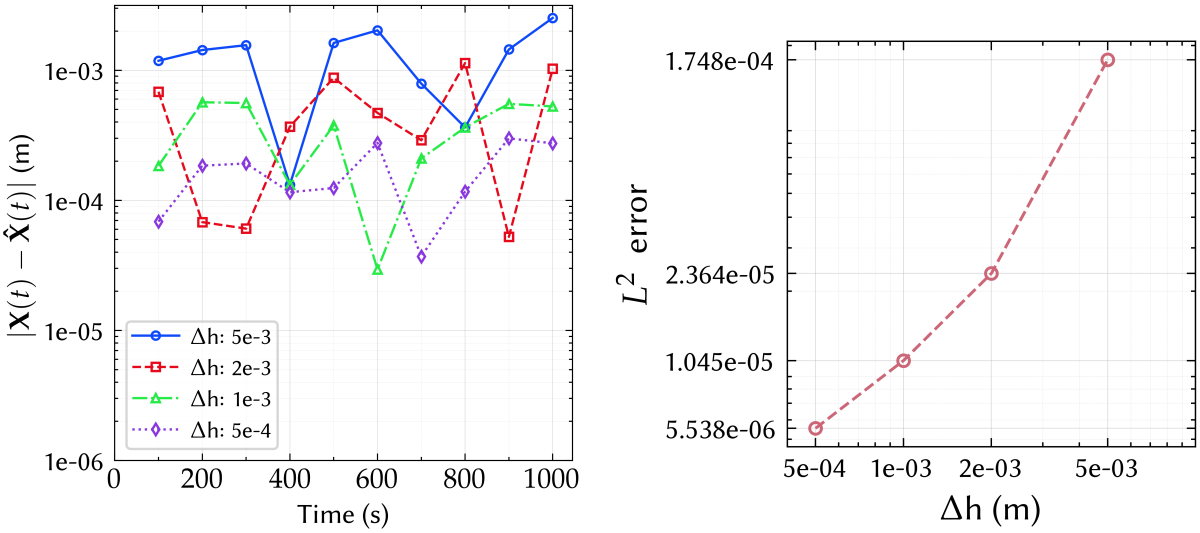


Figure 2: Location of the phase change interface



(a) Absolute error in the location of the interface

(b) Temperature profile convergence analysis

Figure 3: Error Analysis for the Stefan problem for different mesh sizes

Figure 3 shows the location of the interface for simulations with different cell sizes against the analytical solution. The location of the interface is inferred using the volume fraction of the liquid and solid phases present along with the temperature results. The numerical values are plotted at time intervals of 100 s and the absolute error in the location of the interface can be seen in Figure 3a. The temperature profiles obtained numerically can also be compared against the analytical temperature profile in order to calculate the L_2 and L_∞ errors, where $L_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(T_i - T_{analytical,i})^2}$ and $L_\infty = \max_i |T_i - T_{analytical,i}|$ and evaluate the convergence rate.

The errors in the temperature profiles for the various mesh sizes are plotted in Figure 3b. It can be seen that the simulation results converge to the analytical solution with decreasing mesh size.

4.2 Rayleigh-Taylor Instability

The Rayleigh-Taylor (RT) instability occurs when a denser fluid is accelerated into a lighter one, leading to interfacial perturbations and complex vortex formation [54]. This phenomenon, known as the Rayleigh-Taylor (RT) instability, is fundamental to many natural and engineering systems. Although the ideal, undisturbed RT instability is rarely encountered in practice, its essential features arise across a broad range of spatial and temporal scales - where the basic mechanism can be influenced by viscosity, surface tension, compressibility, external accelerations, or stratification[55]

Perturbation growth is driven by oppositely directed gradients of pressure and density at the interface. The density contrast between the two fluids is represented by the non-dimensional *Atwood number*, $At = (\rho_H - \rho_L)/(\rho_H + \rho_L)$, where ρ_H and ρ_L denote the densities of the upper (heavy) and lower (light) fluids respectively. The Atwood number quantifies the strength of the instability, with larger values corresponding to greater density contrasts.

The nonlinear development of the RT instability is central to a wide class of interfacial flows where one fluid penetrates or displaces another [56]. Accurately simulating RT instability serves as a classical benchmark for two-phase flow solvers, requiring detailed computation of interface dynamics and rigorous implementation of interphase interaction terms as discussed in [chapter stuf], and is widely used to validate numerical methods for multiphase flows [57].

Considering a region, $\Omega \in \mathbb{R}^2$, where two immiscible, viscous, and incompressible fluids occupy the regions Ω_H and Ω_L, such that $\Omega = \Omega_H \cup \Omega_L$. The regions $\Omega_{H,L}$ are defined using a scalar height function $h(x,t)$ that graphs the vertical position of the interface at each point along the x-axis.

$$\Omega_i(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : (-1)^i (y - h(x,t)) > 0, t \geq 0\}, \;\; i = H, L$$

The interface, $X(t)$, separating the two immiscible fluids is given as

$$X(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : y = h(x,t), \; t \geq 0\}.$$

With the initial position of the interface denoted by $X(t = t_0) = X_0$, and the initial velocity field by $\underline{\mathbf{v}}(t = t_0) = \underline{\mathbf{v}}_0$, the flow is governed by the incompressible Navier-Stokes equations, surface tension σ, and acceleration $\underline{\mathbf{g}}$. The jump conditions at the interface preserve the velocity field and relate the normal stresses at the interface to the mean curvature.

If density ρ can be assumed to be $\rho_{H,L}$ or the mixture density in cases of the two-fluid and VOF methods respectively. The same can be extended to the velocity, $\underline{\mathbf{v}}$ and the dynamic viscosity μ.

The system of equations governing this two-phase flow system can be expressed as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \underline{\mathbf{v}} = 0 \qquad \text{in } \Omega_i(t) \tag{4.4}$$

$$\frac{\partial (\rho \underline{\mathbf{v}})}{\partial t} + \nabla (\rho \underline{\mathbf{v}} \otimes \underline{\mathbf{v}}) = -\nabla p + \mu \nabla^2 \underline{\mathbf{v}} + \rho \underline{\mathbf{g}} \quad \text{in } \Omega_i(t) \tag{4.5}$$

$$\nabla \cdot \underline{\mathbf{v}} = 0 \tag{4.6}$$

$$[[\underline{\mathbf{v}}]] = 0 \tag{4.7}$$

$$[[(\mu(\nabla \underline{\mathbf{v}} + (\nabla \underline{\mathbf{v}})^T) - (p + \rho \underline{\mathbf{g}})\underline{\underline{\mathbb{I}}})\, \underline{\mathbf{n}}]] = \sigma \kappa \underline{\mathbf{n}} + [[\rho]] \underline{\mathbf{g}} \tag{4.8}$$

$$\underline{\mathbf{v}}(\underline{\mathbf{x}}, t_0) = \underline{\mathbf{v}}_0 \tag{4.9}$$

$$X(t_0) = X_0 \tag{4.10}$$

where $\underline{\mathbf{n}}$ is the normal vector to the interface, p is the pressure field, $\underline{\underline{\mathbb{I}}}$ is the second-order identity tensor, and κ in this case denotes the mean curvature of the interface.

In order to simulate a single mode perturbance to the interface, the initial position of the interface is defined using the amplitude, $A = 0.1 L_x$, and wavelength, $\lambda = L_x$ as follows

$$X_0 = 2L_x + A \cos\left(\frac{2\pi x}{L_x}\right)$$

The dimension L_x is the characteristic length which is chosen to be the width of the simulated domain.

The numerical simulations assume turbulent and unsteady flow. The computational domain is a two-dimensional rectangle, $[0, L_x] \times [0, 4L_x]$ that is discretised using uniform orthogonal quadrilateral cells. Convective and diffusive terms are discretized by the second-order upwind scheme; cell-center gradients are interpolated to face centers using the least-squares method. Interface capturing is performed with the volume-of-fluid (VOF) method using the first-order HRIC scheme. The transient term is approximated using the first-order implicit Euler method. Turbulence effects are modeled using the Shear Stress Transport (SST) k-ω [better word?] closure, and the surface tension force is treated semi-implicitly. Gravitational acceleration acts downward, $g = 1.0\,\mathrm{m/s^2}$. Periodic boundary conditions are imposed in the horizontal direction and slip walls at the top and bottom.

The DNS study undertaken in [58] is taken to be the canonical benchmark and the physical parameters are chosen to maintain dynamic similarity.

According to [59], a reasonable solution to the RT instability with $At \approx 0.75$ can be obtained with a 128×512 mesh. With similar mesh resolutions being used in [58, 60] to study similar RT instabilities, the domain is discretised using a uniform orthogonal quadrilateral mesh with size $\Delta h = 4.5 \times 10^{-3}$ m. Additional mesh sizes are evaluated in order to study mesh convergence. Time is non-dimensionalised using the Atwood number and the timestep for the simulations, similar to [61, 58], is selected as $\Delta t = 2.5 \times 10^{-4}/\sqrt{At}$ s.

The physical parameters and the simulation setup, along with the various cases studied are summarised in Tables 2 and 3.

Spatial and Temporal Discretisation	
Domain	2D, $[0, L_x] \times [0, 4L_x]$, where $L_x = 1$ m
Time discretization	Implicit unsteady, 1st order
Time step, Δt	3.5×10^{-3} s
Physical Properties	
Amplitude of perturbation, A	0.1
Wavelength of perturbation, λ	$L_x = 1.0$ m
Surface tension, σ	1.0×10^{-5} N·m^{-1}
Gravity, $\underline{\mathbf{g}}$	$[0, -1.0]$ m·s^{-2}
Numerical Methods	
Multiphase model	VOF with HRIC convection
Flow model	Turbulent, SST k–ω
Spatial discretization	2nd order (flow), 1st order (HRIC)
Gradient reconstruction	Least-squares

Table 2: CFD simulation parameters for Rayleigh-Taylor instability study

	At = 0.2			At = 0.5			High At		
Parameter	Re300	Re1000	Re3000	Re300	Re1000	Re3000	At30	At1000	At1500
ρ_H [kg·m^{-3}]	1.5	1.5	1.5	3.0	3.0	3.0	3.0	1.0	1.5
ρ_L [kg·m^{-3}]	1.0	1.0	1.0	1.0	1.0	1.0	0.1	0.001	0.001
$\mu_{H,L}$ [Pa·s]	0.005	0.0015	0.0005	0.01	0.003	0.001	0.001	0.001	0.001
Mesh size Δh [m]	0.0045	0.0045	0.0045	(0.01, 0.007, 0.0045)	0.0045	(0.01, 0.007, 0.0045)	0.0045	0.0045	0.0045

Table 3: Fluid properties and mesh resolutions for all simulation cases

Figure 4 depicts the evolution of the RT instability for Reynolds numbers 300, 1000, and 3000. The initial evolution of the interface is through exponential growth of the infinitesimal disturbance of the interface [62]. The growth follows linear stability theory which transitions into increasing non-linear behaviour as the amplitude grows to orders of 0.1λ [55]. This non-linear evolution is marked by the development of the distinct "bubbles" of lighter fluid rising as the "spikes" of the heavier fluid fall down. The evolving interface experiences increasing vorticity and the increasing velocity shear results in the development of Kelvin-Helmholtz type instabilities [58, 56] leading to the roll-up of the spikes as the amplitude grows to

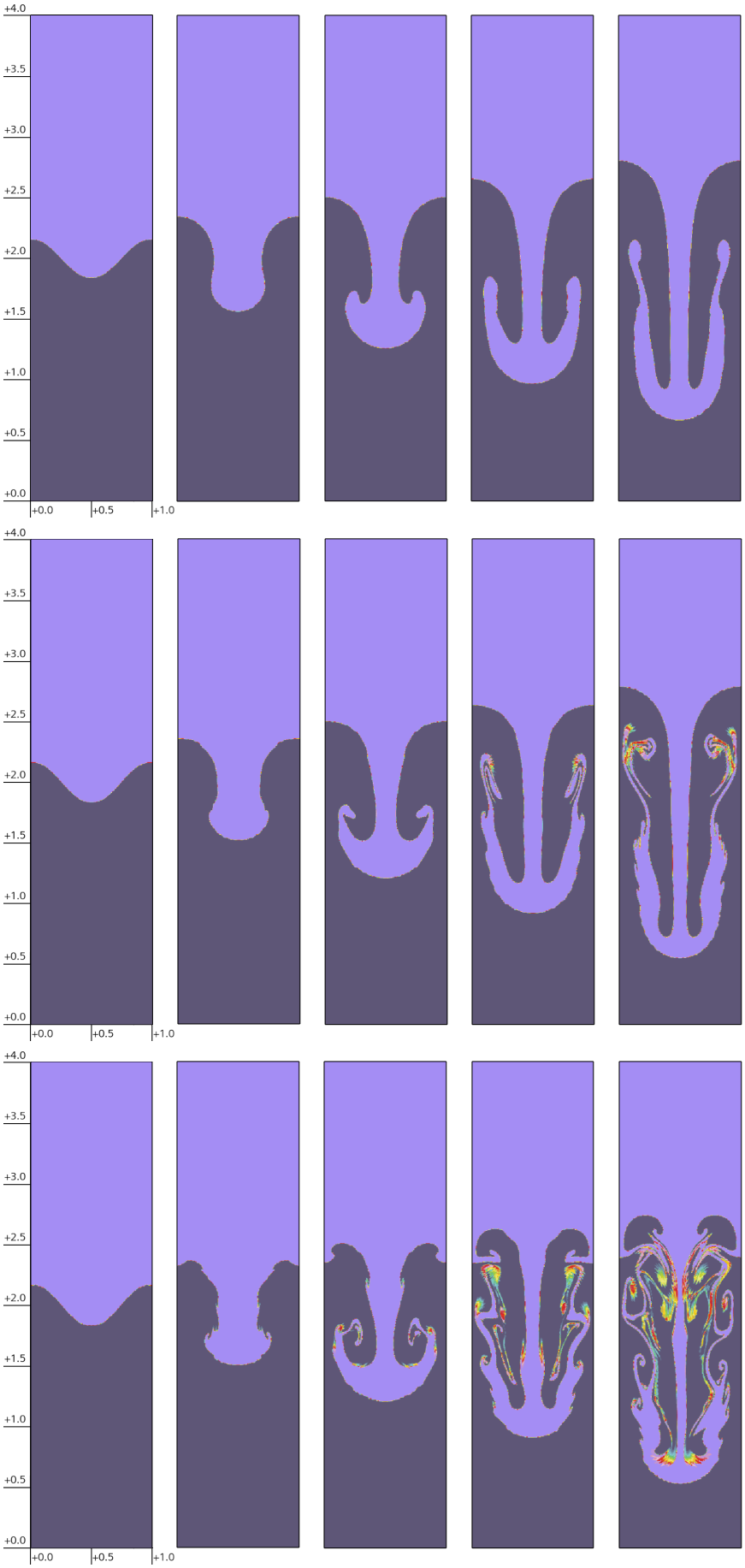


Figure 4: Evolution of Rayleigh-Taylor instability for Atwood number At = 0.5 at different Reynolds numbers.

the order of $0.5 - 1\lambda$. For the case of $Re = 3000$, these instabilities are noticeable much earlier all along the interface, leading to a secondary perturbation along the boundary of the domain. Further growth leads to secondary vortices along the spike ends which leads to chaotic mixing and complex flow features. [55].

- **Effect of Reynolds and Atwood Number:** Increasing the Reynolds number has a relatively small effect on the penetration lengths of the spikes and bubbles - with $Re = 300$ showing the least penetration depths. However, it does have a significant effect on the topology of the interface, with increasingly chaotic vortices visible at the spike rollup and the ensuing turbulent mixing. At $Re = 300$, it is seen in Figure 4, that the flow remains relatively laminar with smooth interfaces, whereas $Re = 3000$ shows significant turbulent mixing and vortical complex structures. High At-

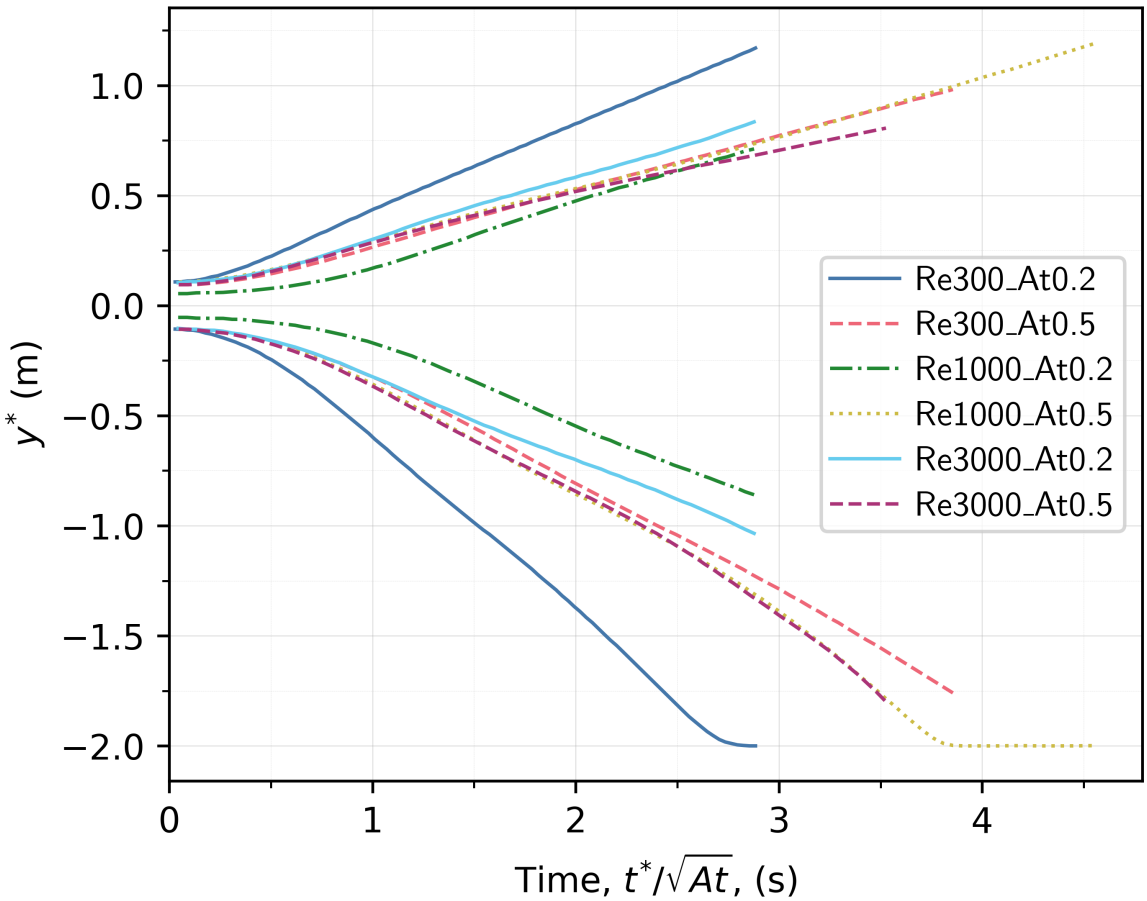
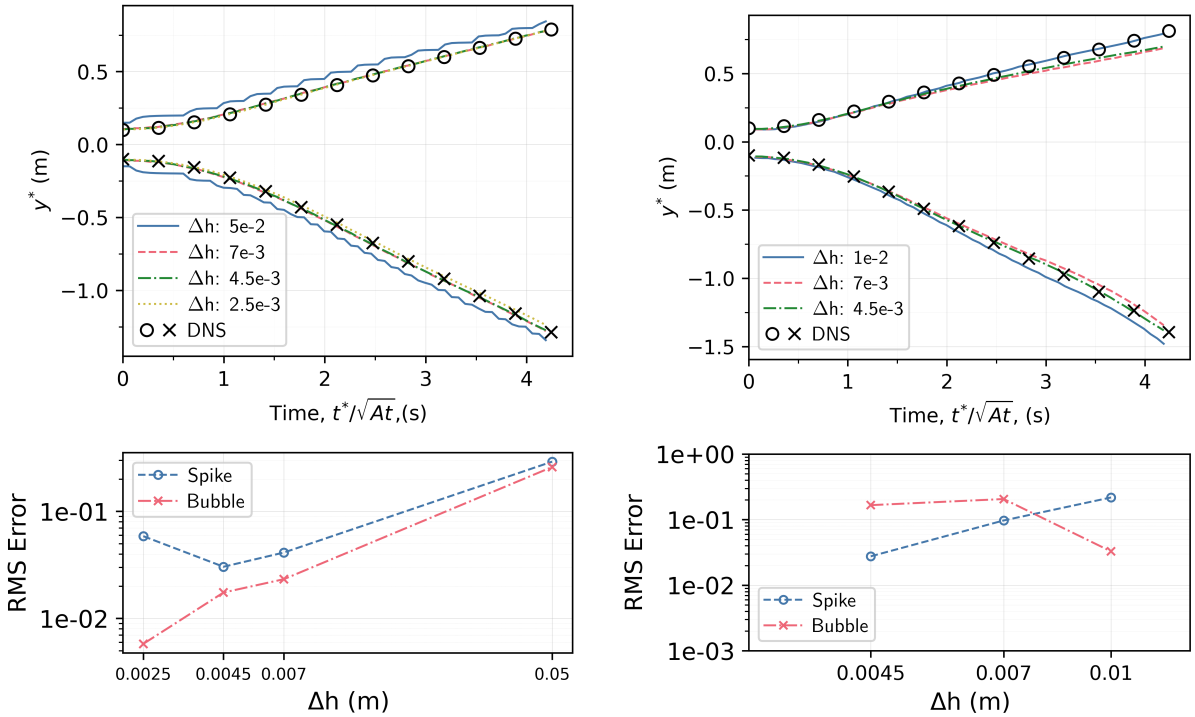


Figure 5: Evolution of the RT instability with varying Reynolds and Atwood numbers

wood numbers promote non-linearity and increased turbulent mixing, and affect the initial interface growth. The effect and prominence of the counter-rotating vortices at the tip of the spike decreases and reduces the chaotic flow features at the tips of the secondary vortices. The spike also falls much quicker. This is due to the density-driven acceleration outweighing the viscous effects as is seen in Figure 8 and in the very high density ratio simulations of the RT instability in [60].

- **Grid Convergence:** In order to evaluate the quality of the mesh, the maximum vertical distance of the spike and bubble tip as measured from the undisturbed initial interface, $y^* = |y_{s,b} - y_0|$, are evaluated for Reynolds numbers, $Re = 300, 3000$, and Atwood number, $At = 0.5$. Locations of the spike and bubble tips are recorded at every time-step and extrapolated in order to compare to the DNS data available from [58].

Figure 6 shows the comparison the results of the simulations with the DNS location of the bubble and spike tips along with the root mean square defined as $err = \sqrt{\sum_{t_0}^{t_f}(y_b - y_{dns})^2}$. The RMS errors and the maximum absolute errors are further summarized in table [table].



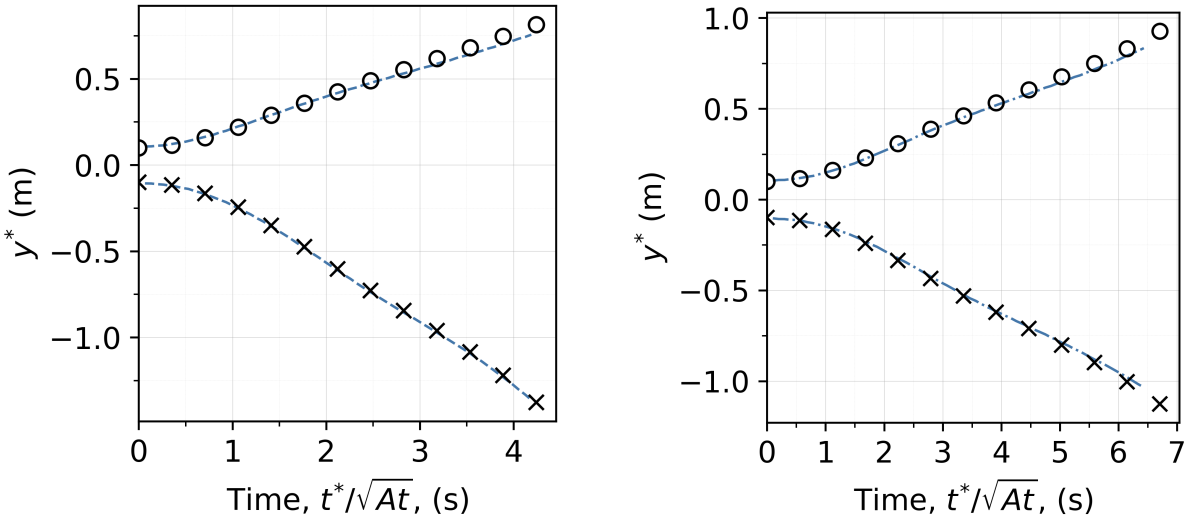
(a) $Re = 300$ **(b)** $Re = 3000$

Figure 6: Grid convergence analysis for Rayleigh-Taylor instability

Figure 7 compares the RT instabilities for $Re = 1000, 3000$ and $At = 0.5, 0.2$ respectively without the evaluation of the errors.

Although, Star-CCM+, qualitatively, accurately simulates the RT instability, the calculated RMS and absolute errors in the bubble and spike tip locations show erratic behaviour with decreasing mesh size. For $Re = 300$, the sudden increase in the RMS error at $\Delta h = 0.0025$ can be a result of [?] For higher Reynolds numbers, the bubble tip distances are consistently under-predicted for $Re = 1000, 3000$ at later simulation times. For $Re = 3000$, at later simulation times, secondary perturbations develop at the domain edges. These secondary perturbations develop into secondary spikes that interfere with the turbulent vortices at the spike tips and the main spike stem. This is, interestingly, not seen for the coarsest mesh size for $Re = 3000$. This could be developing Kelvin-Helmholz instabilities at the surface that transition into secondary RT instabilities [63]. Although Kelvin-Helmholz instabilities develop along the interface at higher Reynolds number, this secondary perturbation is first observed for $Re = 10000$ in [58].

This can be ascribed to either the lower order scheme used in the Star-CCM+ simulations or the choice of turbulence modelling using eddy viscosity models instead of completely resolving turbulence. For Atwood number that are not close to zero, the Boussinesq approximation is also invalid which is an integral assumption in RANS turbulence models. Star-CCM+, in order to accommodate the spurious currents that may be introduced through the use of eddy viscosity models, provides models that aid in the damping of momentum and/or turbulence at the interface, however, the addition of these models leads to significant deviations from the DNS results.



(a) $Re = 1000$, $At = 0.5$ **(b)** $Re = 3000$, $At = 0.2$

Figure 7: Comparison with DNS results for different Reynolds numbers and Atwood numbers

4.3 Liquid Infiltration of Evaporation Chamber

The evaporation chamber and the nozzle are modelled together as a 2D domain as shown in Figure [FIG]. Star-CCM+, during the creation of a 2D domain, prescribes an automatic, unchangeable depth of 1 m to the model. In order to maintain dynamic similarity, the mass flow rate is compared between the 3D experiment and the 2D model. To this end, the inlet velocity is kept similar in the two experiments. It is assumed that approximately 150 mL water is discharged into the evaporation chamber during the experiment.

The diameter of the inlet, $d_{in} = 0.017m$, is taken to be the reference length. Assuming a discharge time $t_{dis} = 10$ s and a constant flow rate, the inflow velocity of water in the experiment can be calculated as

$$\dot{Q} = Q/t_{dis} = 0.00015/10 = v\left(\pi d_{in}^2/4\right) \tag{4.11}$$

The flow rate is assumed to be constant for the discharge time. The inlet velocity is thus dependent on the discharge time as $v = 0.00015/\left(d_{in}\, t_{dis}\right)$. The gravitational acceleration is assumed to be $\underline{\mathbf{g}} = 0.4\,\mathrm{m{\cdot}s^{-2}}$ along the negative y-axis.

In the actual 3D experiment, the height of the water at the end of the injection under steady state conditions can be estimated using the dimensions and the volume of the discharged water to ≈ 42 mm. The water injection is similarly cutoff, at around 2.75 s, after the water level reaches approximately 42 mm in the simulation, following which the simulation proceeds to run for another 5 s with no inflow velocity.

Section probes present along the mid-section of the evaporation chamber and at various points along the nozzle measure the temperature as they would be measured in the actual experiment. An isosurface defines the water-air interface where the volume fraction of water is 0.5. This isosurface marks the highest position of the water along the mid-section probe and records the $y-$axis velocity at the point.

- **Spatial and Temporal Convergence:** In order to assess the spatial convergence of the solution, four uniform quadrilateral meshes are used. Using the reference length, L_x, the mesh sizes from coarse to fine are $\Delta h_1 = d_{in}/4$, $\Delta h_2 = d_{in}/8$, $\Delta h_3 = d_{in}/16$, and $\Delta h_4 = d_{in}/32$ respectively. A uniform timestep, $\Delta t = 0.001$ s is used for all the meshes along with the same initial conditions. Data is recorded at every 0.01 s. The maximum height of the isosurface against the middle section is used to measure the water level. The velocities at various points along the middle section of the chamber are also recorded. Figure 9 depicts the recorded water level against simulation time

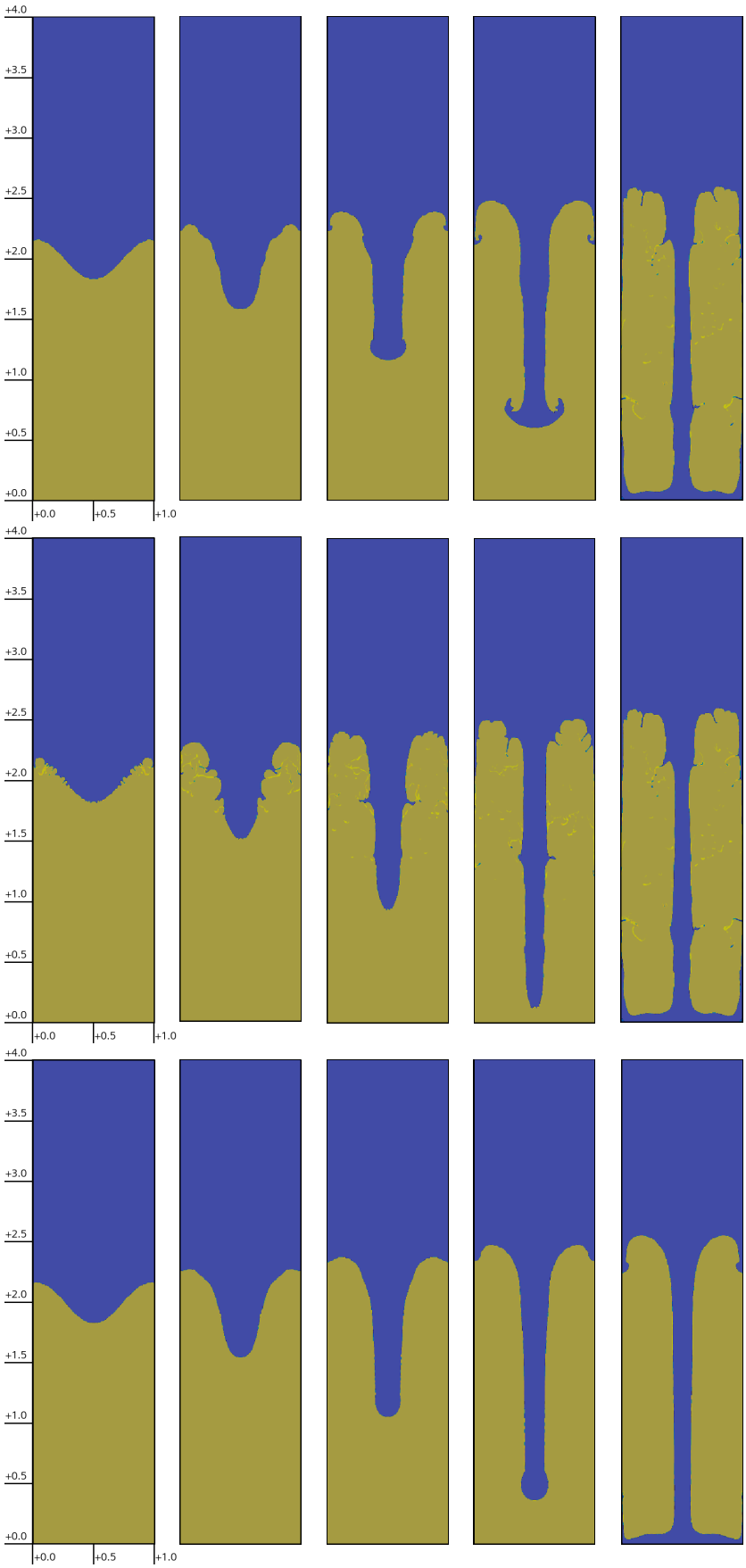


Figure 8: Evolution of Rayleigh-Taylor instability for high Atwood numbers At = 30, 1000, and 1500.

and the y-axis component of the velocities measured along the middle section at the simulation stop time, t_f. All quantities have been non-dimensionalised using the reference length, velocity, and time scales. The coarsest mesh size, h_1, tends to behave like a vibrating membrane with

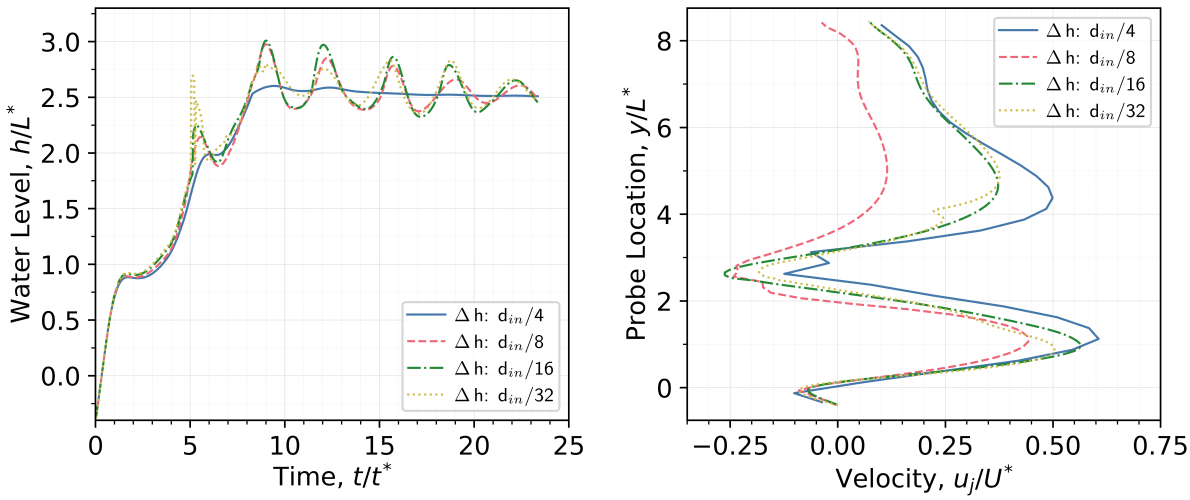


Figure 9: Water level and y-axis velocity component at time, t_f, for various mesh refinements

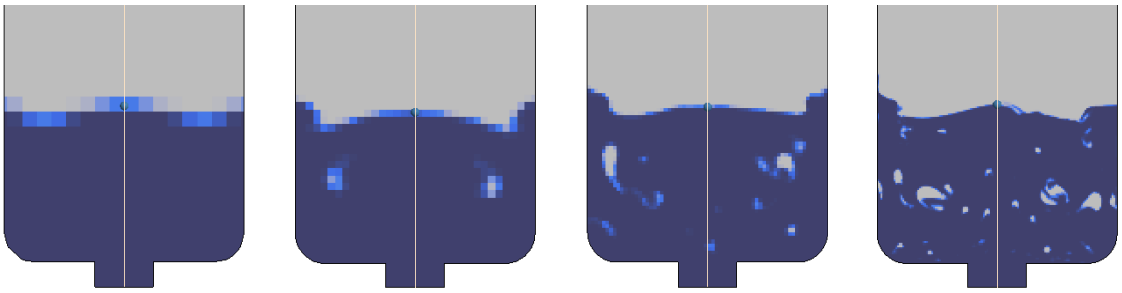


Figure 10: Interface at time, $t = 3.25\,\mathrm{s}$

no discernible development of any waves. The fluid bulk does not contain any air pockets as the mesh is too coarse to resolve any fine details. The finer mesh, h_2, is able to resolve bubbles in the fluid bulk during times of low bubble velocities and when the bubble diameter spans atleast four cells. The coarse mesh is, however, able to resolve waves and the oscillatory motion of the isosurface is clearly noticeable. The finest mesh size h_4 is able to resolve merging and breaking apart of the bubbles, droplets escaping the bulk fluid, as well as the flow features developed due to the interaction between the phases at the free surface. The mesh h_3 while also resolving the bubble interactions in the bulk flow and some coarse droplets, is unable to resolve the interface as sharply. However, the increasing void fraction is apparent with decreasing mesh size.

- [ON EVOLUTION OF INTERFACE and FLOW FIELD and RESULTING HEIGHT OF WATER]
- **Comments:** Solution is highly sensitive; development is mostly symmetric; the thinning of the interface before separating into droplets is called 'pinching off'; that peak in water level is higher and also thinner which is nice.

References

[1] Mamoru Ishii and Takashi Hibiki. *Thermo-Fluid Dynamics of Two-Phase Flow*. Springer New York, 2011.

[2] George Yadigaroglu and Gad Hetsroni. Nature of Multiphase Flows and Basic Concepts. In George Yadigaroglu and Geoffrey F. Hewitt, editors, *Introduction to Multiphase Flow: Basic Concepts, Applications and Modelling*, pages 1–37. Springer International Publishing.

[3] Thomas J Hanratty. Physics of Gas–Liquid Flows.

[4] Daniel Wickert and Günther Prokop. Simulation of water evaporation under natural conditions—A state-of-the-art overview. 3(4):242–249.

[5] Yixiang Liao and Dirk Lucas. Computational modelling of flash boiling flows: A literature survey. 111:246–265, 2017.

[6] Eran Sher, Tali Bar-Kohany, and Alexander Rashkovan. Flash-boiling atomization. 34(4):417–439.

[7] T H Cochran and J C Aydelott. Effects of fluid properties and gravity level on boiling in the discrete bubble region.

[8] Jeongmin Lee, Issam Mudawar, Mohammad M. Hasan, Henry K. Nahra, and Jeffrey R. Mackey. Experimental and computational investigation of flow boiling in microgravity. 183:122237.

[9] D. M. Qiu, V. K. Dhir, D. Chao, M. M. Hasan, E. Neumann, G. Yee, and A. Birchenough. Single-Bubble Dynamics During Pool Boiling Under Low Gravity Conditions. 16(3):336–345.

[10] Chun Long Xu, Yong Chao Shi, Xiao Xing Jin, Zuo Sheng Lei, Yun Bo Zhong, and Jia Hong Guo. Numerical Study of Marangoni Convection around a Single Vapor Bubble during Pool Boiling. 624:262–266.

[11] Vladimir Serdyukov, Ivan Malakhov, and Anton Surtaev. The Influence of Pressure on Local Heat Transfer Rate under the Vapor Bubbles during Pool Boiling. 16(9):3918.

[12] Hui Zhang, Issam Mudawar, and Mohammad M. Hasan. Experimental and theoretical study of orientation effects on flow boiling CHF. 45(22):4463–4477.

[13] (PDF) A Method for Assessing the Importance of Body Force on Flow Boiling CHF.

[14] D. Saury, S. Harmand, and M. Siroux. Experimental study of flash evaporation of a water film. 45(16):3447–3457.

[15] Haoyu Wu, Weimin Liu, Xiaoming Li, Fengyun Chen, and Longbin Yang. Experimental Study on Flash Evaporation under Low-pressure Conditions. 22(2):213–220.

[16] Yixiang Liao and Dirk Lucas. A review on numerical modelling of flashing flow with application to nuclear safety analysis. 182:116002, 2021.

[17] Kazuma Tani, Seiya Tanaka, and Takehiro Himeno. NUMERICAL SIMULATION OF SINGLE BUBBLE GROWTH DURING DEPRESSURIZATION.

[18] Sami Mutair and Yasuyuki Ikegami. Experimental investigation on the characteristics of flash evaporation from superheated water jets for desalination. 251(1):103–111.

[19] S. Vahaji, A. A. Akbarzadeh, A. Date, S. C. P. Cheung, and J. Y. Tu. Efficiency of a two-phase nozzle for geothermal power generation. 73(1):229–237.

[20] O. Miyatake, T. Tomimura, Y. Ide, and T. Fujii. An experimental study of spray flash evaporation. 36(2):113–128.

[21] O. Miyatake, T. Tomimura, Y. Ide, M. Yuda, and T. Fujii. Effect of liquid temperature on spray flash evaporation. 37(3):351–366.

[22] S. Gopalakrishna, V. M. Purushothaman, and N. Lior. An experimental study of flash evaporation from liquid pools. 65:139–151.

[23] R. J. Peterson, S. S. Grewal, and M. M. El-Wakil. Investigations of liquid-flashing and evaporation due to sudden depressurization. 27(2):301–310.

[24] D. Saury, S. Harmand, and M. Siroux. Flash evaporation from a water pool: Influence of the liquid height and of the depressurization rate. 44(10):953–965, 2005.

[25] Hahne, E., and G. Barthau (2000) Evaporation waves in flashing process, - Google Search.

[26] P Reinke and G Yadigaroglu. Explosive vaporization of superheated liquids by boiling fronts. 27(9):1487–1516.

[27] Haoyu Wu, Weimin Liu, Xiaoming Li, Fengyun Chen, and Longbin Yang. Experimental Study on Flash Evaporation under Low-pressure Conditions. 22(2):213–220.

[28] Yixiang Liao, Dirk Lucas, Eckhard Krepper, and Roland Rzehak. Flashing evaporation under different pressure levels. 265:801–813, 2013.

[29] Sanjeev Kumar, Marc Medale, Paolo Di Marco, and David Brutin. Sessile volatile drop evaporation under microgravity. 6(1):37.

[30] M Wörner. A Compact Introduction to the Numerical Modeling of Multiphase Flows.

[31] Andrea Prosperetti and Gretar Tryggvason. COMPUTATIONAL METHODS FOR MULTIPHASE FLOW.

[32] Christopher E. Brennen. *Fundamentals of Multiphase Flow*. Cambridge University Press, 1. publ., repr edition.

[33] Gretar Tryggvason, Ruben Scardovelli, and Stéphane Zaleski. *Direct Numerical Simulations of Gas-Liquid Multiphase Flows*. Cambridge Univ. Press, 1. publ edition, 2011.

[34] Vinesh H. Gada, Mohit P. Tandon, Jebin Elias, Roman Vikulov, and Simon Lo. A large scale interface multi-fluid model for simulating multiphase flows. 44:189–204.

[35] Mario F. Trujillo. Reexamining the one-fluid formulation for two-phase flows. 141:103672.

[36] S. Hardt and F. Wondra. Evaporation model for interfacial flows based on a continuum-field representation of the source terms. 227(11):5871–5895.

[37] Lindsey Anne Gilman. Development of a general purpose subgrid wall boiling model from improved physical understanding for use in computational fluid dynamics.

[38] Lars Davidson. An Introduction to Turbulence Models.

[39] D. C. Besnard and F. H. Harlow. Turbulence in multiphase flow. 14(6):679–699.

[40] Ch Hirsch. *Numerical Computation of Internal and External Flows: Fundamentals of Computational Fluid Dynamics*. Elsevier/Butterworth-Heinemann, 2nd ed edition, 2007.

[41] Stephen B. Pope. *Turbulent Flows*. Cambridge Univ. Press, 1. publ., 12. print edition.

[42] David C. Wilcox. *Turbulence Modeling for CFD*. DCW Industries, 3rd ed edition, 2006.

[43] F. R. Menter. Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA Journal*, 32(8):1598–1605, 1994.

[44] Rosemary M Sugrue. A Robust Momentum Closure Approach for Multiphase Computational Fluid Dynamics Applications.

[45] Eckhard Krepper, Roland Rzehak, and Dirk Lucas. Validation of a closure model framework for turbulent bubbly two-phase flow in different flow situations. 340:388–404.

[46] Rosemary Sugrue, Ben Magolan, Nazar Lubchenko, and Emilio Baglietto. Assessment of a simplified set of momentum closure relations for low volume fraction regimes in STAR-CCM+ and OpenFOAM. 110:79–87.

[47] Yixiang Liao, Tian Ma, Liu Liu, Thomas Ziegenhein, Eckhard Krepper, and Dirk Lucas. Eulerian modelling of turbulent bubbly flow based on a baseline closure concept. 337:450–459.

[48] M Colombo, R Rzehak, M Fairweather, Y Liao, and D Lucas. Benchmarking of CFD Modelling Closures for Two-Phase Turbulent Bubbly Flows.

[49] Sibel Tas-Koehler, Yixiang Liao, and Uwe Hampel. A critical analysis of drag force modelling for disperse gas-liquid flow in a pipe with an obstacle. 246:117007.

[50] Shumon Koga and Miroslav Krstic. Phase change model: Stefan problem. In *Materials Phase Change PDE Control & Estimation: From Additive Manufacturing to Polar Ice*, pages 1–13. Springer International Publishing.

[51] Tobias Jonsson. On the one dimensional Stefan problem.

[52] Leonardo Boledi, Benjamin Terschanski, Stefanie Elgeti, and Julia Kowalski. A level-set based space-time finite element approach to the modelling of solidification and melting processes. 457:111047.

[53] Daniele Andreucci. Lecture notes on the Stefan problem.

[54] G Birkhoff. TAYLOR INSTABILITY AND LAMINAR MIXING.

[55] D.H. Sharp. An overview of Rayleigh-Taylor instability. 12(1):3–18.

[56] Grétar Tryggvason. Numerical simulations of the Rayleigh-Taylor instability. 75(2):253–282.

[57] Jason Goulding, Mehrnaz Ayazi, Tamar Shinar, and Craig Schroeder. Simplified conservative discretization of the Cahn-Hilliard-Navier-Stokes equations. 519:113382.

[58] Arash Hamzehloo, Paul Bartholomew, and Sylvain Laizet. Direct numerical simulations of incompressible Rayleigh-Taylor instabilities at low and medium Atwood numbers. 33(5):054114, 2021.

[59] Shahab Mirjalili, Christopher B. Ivey, and Ali Mani. Comparison between the diffuse interface and volume of fluid methods for simulating two-phase flows. 116:221–238, 2019.

[60] Ziyang Huang, Guang Lin, and Arezoo M. Ardekani. Consistent, essentially conservative and balanced-force Phase-Field method to model incompressible two-phase flows. 406:109192, 2020.

[61] Hang Ding, Peter D.M. Spelt, and Chang Shu. Diffuse interface model for incompressible two-phase flows with large density ratios. 226(2):2078–2095, 2007.

[62] Malcolm J. Andrews and Stuart B. Dalziel. Small Atwood number Rayleigh–Taylor experiments. 368(1916):1663–1679.

[63] Feng Chen, Aiguo Xu, Yudong Zhang, Yanbiao Gan, Bingbing Liu, and Shuang Wang. Effects of the initial perturbations on the Rayleigh-Taylor-Kelvin-Helmholtz instability system. 17(3):335005.

[64] C. M. Usiskin and R. Siegel. An experimental study of boiling in reduced and zero gravity fields. *Journal of Heat Transfer*, 83(3):243–253, 1961. NASA Technical Report.

[65] M. M. Hasan, C. S. Lin, R. H. Knoll, and M. D. Bentz. Nucleate pool boiling in the long duration low gravity environment of the Space Shuttle. *NASA Technical Memorandum*, (106049), 1993.

[66] J. Stengler, T. Weith, J. Clade, and T. Horstmann. Experimental study on low temperature desalination by flash evaporation. *Desalination and Water Treatment*, 106:144–151, 2018.

[67] L. Zhang, H. Wang, and M. Chen. Numerical study of growth of a vapor bubble in superheated liquid with salt accumulation. *Heat Transfer Engineering*, 45(8):672–685, 2024.

[68] S. T. Hong, C. Dang, and E. Hihara. Review on state-of-the-art research in pool and flow boiling under microgravity. *International Journal of Heat and Mass Transfer*, 200:123456, 2023.

[69] J. Straub. Boiling heat transfer and bubble dynamics in microgravity. *Advances in Space Research*, 28:629–640, 2001.

[70] J. L. Ribeiro Jr et al. Experimental study on bubble size distributions in a direct-contact evaporator. *Brazilian Journal of Chemical Engineering*, 21:117–127, 2004.

[71] M. Oka and J. Kim. Gravity influence on heat transfer rate in flow boiling. *Microgravity Science and Technology*, 24:161–167, 2012.

[72] K. Zhang et al. Bubble growth during subcooled nucleate boiling on a vertical heater. *Nuclear Engineering and Design*, 323:180–191, 2017.

[73] E. Elele et al. Single-bubble water boiling on small heater under earth's gravity and low gravity. *Nature Communications*, 9:3842, 2018.

[74] I. Mudawar et al. Iss flow boiling and condensation experiment campaign 1 summary. Technical Report NASA/TP-20230002447, NASA Glenn Research Center, 2022.

[75] C. Chi et al. Experimental results for a novel flash evaporation and desalination system. *ASME Journal of Heat Transfer*, 145:041601, 2023.

[76] A. Sarma et al. Flash evaporation of refrigerant r134a through nozzles. *Heat Transfer Engineering*, 45:934–950, 2024.

[77] R. Siegel and E. G. Keshock. Effects of reduced gravity on nucleate boiling bubble dynamics in saturated water. *AIChE Journal*, 10:509–517, 1964.

[78] L. Hao. Analysis of bubble growth and motion dynamics in superheated liquid during flash evaporation. *International Journal of Heat and Mass Transfer*, 151:119356, 2020.

[79] S. Banerjee. Nucleate boiling under different gravity values: numerical simulations and data-driven techniques. 2022.

[80] V. Serdyukov, I. Malakhov, and A. Surtaev. The influence of pressure on local heat transfer rate under the vapor bubbles during pool boiling. *Energies*, 16:3918, 2023.

[81] Jan Wilhelm Gärtner. Modeling and simulation of flash evaporation of cryogenic liquid jets, 2024.

[82] Ahmad Mansour and Norbert Müller. A review of flash evaporation phenomena and resulting shock waves. 107:146–168, 2019.

[83] Osamu Miyatake, Itsuo Tanaka, and Noam Lior. A simple universal equation for bubble growth in pure liquids and binary solutions with a nonvolatile solute. 40(7):1577–1584, 1997.

[84] Quang Dang Le, Riccardo Mereu, Giorgio Besagni, Vincenzo Dossena, and Fabio Inzoli. Computational Fluid Dynamics Modeling of Flashing Flow in Convergent-Divergent Nozzle. 140(10):101102, 2018.

[85] Volume-Averaged Macroscopic Transport Equations. In E. Blosch, H. Ouyang, J. Liu, S. S. Thakur, and Wei Shyy, editors, *Computational Techniques for Complex Transport Phenomena*, pages 231–259. Cambridge University Press, 1997.

[86] S. Mostafa Ghiaasiaan. *Two-Phase Flow, Boiling and Condensation: In Conventional and Miniature Systems*. Cambridge University Press, 2007.

[87] Geoffrey F. Hewitt and George Yadigaroglu. Modelling Strategies and Two-Phase Flow Models. In George Yadigaroglu and Geoffrey F. Hewitt, editors, *Introduction to Multiphase Flow: Basic Concepts, Applications and Modelling*, pages 39–77. Springer International Publishing, 2018.

[88] George Yadigaroglu. Interfacial Instabilities. In George Yadigaroglu and Geoffrey F. Hewitt, editors, *Introduction to Multiphase Flow: Basic Concepts, Applications and Modelling*, pages 79–94. Springer International Publishing, 2018.

[89] C. Truesdell and K. R. Rajagopal. *An Introduction to the Mechanics of Fluids*. Birkhäuser, 2000.

[90] Michel O. Deville and Thomas B. Gatski. *Mathematical Modeling for Complex Fluids and Flows*. Springer Berlin Heidelberg, 2012.

[91] Maxwell Hutchinson. A DISSERTATION SUBMITTED TO THE FACULTY OF THE DIVISION OF THE PHYSICAL SCIENCES IN CANDIDACY FOR THE DEGREE OF DOCTOR OF PHILOSOPHY.

[92] S. Dartevelle. Comprehensive Approaches to Multiphase Flows in Geophysics - Application to non-isothermal, nonhomogenous, unsteady, large-scale, turbulent dusty clouds I. Hydrodynamic and Thermodynamic RANS and LES Models.

[93] Arindam Banerjee. Rayleigh-Taylor Instability: A Status Review of Experimental Designs and Measurement Diagnostics. 142(12):120801.

[94] Oleg Schilling. Progress on Understanding Rayleigh–Taylor Flow and Mixing Using Synergy Between Simulation, Modeling, and Experiment. 142(12):120802.

[95] Marco Colombo and Michael Fairweather. Influence of multiphase turbulence modelling on interfacial momentum transfer in two-fluid eulerian-eulerian cfd models of bubbly flows. *Chemical Engineering Science*, 195:968–984, 2019.

[96] Wei Shyy, S. S. Thakur, H. Ouyang, J. Liu, and E. Blosch. *Computational Techniques for Complex Transport Phenomena*. Cambridge University Press, 1 edition.

[97] Clifford Truesdell. *A First Course in Rational Continuum Mechanics*. Number v. 71 in Pure and Applied Mathematics. Academic Press, 2nd ed edition.